

Reliability of Power Semiconductor Modules: A State-of-the-Art Review

ENEKO AGIRREZABALA , IOSU AIZPURU , DAVID GARRIDO , AND ANE PORTILLO 

Electronics and Computing Department, Mondragon Unibertsitatea, 20120 Hernani, Spain

CORRESPONDING AUTHOR: ENEKO AGIRREZABALA (e-mail: eagirrezabala@mondragon.edu)

This work was supported by the Department of Education of the Basque Government under the Non Doctoral Research Staff Training Program under Grant PRE_2023_1_0212.

ABSTRACT Power electronic converters are essential in modern energy systems, facilitating efficient energy conversion in applications such as renewable energy, transportation, and industry. Power semiconductors are crucial to ensure reliability within these converters, often representing the weakest link in performance and durability. Failures in these devices can lead to costly downtime, decreased performance, and safety risks in demanding applications. This article provides a comprehensive analysis of the reliability challenges and solutions associated with power semiconductors, specifically focusing on silicon (Si) IGBTs and silicon carbide (SiC) MOSFETs. It examines key failure mechanisms, such as gate-oxide degradation and thermo-mechanical aging, at both the chip and package levels. Additionally, the article offers a market-oriented overview of standard power semiconductor packages and insights into current manufacturing trends and their implications for reliability. It also explores Active Thermal Control (ATC) techniques, such as gate voltage modulation, which can be implemented at the user level to mitigate stress-related failures. The findings intend to inform future researchers and developers, ensuring that power electronics systems can meet the increasing demand for reliable, efficient, and safe energy solutions.

INDEX TERMS Power semiconductors, reliability, Si-IGBT, SiC-MOSFET, chip-level reliability, package-level reliability, active thermal control (ATC), power module classification, market overview.

I. INTRODUCTION

Over the past decades, the growing energy demand (more than 70% worldwide [1]) has positioned electrical energy as one of the key factors in our daily lives. The transition towards renewable energy sources, including wind, solar, and hydropower, is essential in achieving net-zero emissions by 2050 [2]. Power electronics play a crucial role in ensuring the reliability and efficiency of energy conversion systems [3], [4]. The enhancement of these devices directly contributes to the overall efficiency of the energy sector. Power electronics, and especially power converters, are increasingly present in applications where generating, transporting, or consuming electric power efficiently and reliably is required. These applications cover a wide range of areas, from renewable energy generation to its distribution and final use in industry, mobility, and household appliances.

Field data indicate that power electronic converters are often the weakest link in terms of electrical system reliability [5], [6]. In wind farms, these converters are linked to about

13% of failures and contribute to 18% of operational downtime [7], [8]. In utility-scale photovoltaic (PV) installations, inverters are responsible for nearly 37% of all unscheduled maintenance events [9], [10]. In traction applications, as reported by Alstom, while preventive maintenance is not applied to converters, they still account for 38% of the total maintenance costs [11]. The semiconductor switches within these converters are particularly vulnerable to failure [4]. Such failures are often due to unexpected overloads and system transients, as well as thermo-mechanical stresses leading to long-term wear-out [5]. This vulnerability can result in power semiconductor modules failing to meet their expected lifetimes, especially when subjected to overvoltage and overcurrent conditions outside the Safe Operating Area (SOA) [4], [12].

Power electronics systems now need to meet more demanding grid codes and provide essential support to the grid, such as in the case of Low-Voltage Ride-Through (LVRT) events [13], [14]. These systems are often used in remote

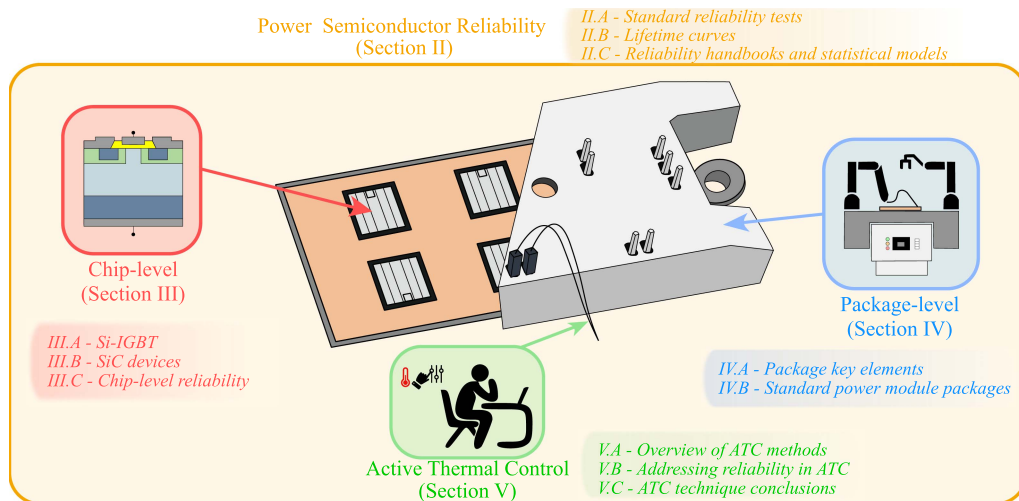


FIGURE 1. Diagram of the paper's structure, highlighting the focus of each section within the power semiconductor module.

locations and must function in extreme environmental conditions, such as high temperatures. Additionally, the emphasis on higher efficiency and power density means that power converters are operating closer to their thermal and electrical limits, which increases the likelihood of stress-related failures [15], [16].

While previous studies have made significant advancements in analysing the reliability of power semiconductors, there are still opportunities to build on these bases and offer a more integrated perspective. For instance, [15] offers a thorough analysis of failures at both the chip and package levels, highlighting key failure mechanisms. However, this work does not address reliability enhancement strategies from the perspective of power electronics designers and end-users, such as EV manufacturers, power grid operators, and fast-charging infrastructure developers, who can actively influence device operation through their design considerations and control strategies. In addition, [17] complements this by exploring reliability assessment strategies and accelerated aging tests for power electronics converters. Along with analysing failure mechanisms, this article includes a valuable discussion of Active Thermal Control (ATC) strategies. However, it does not provide an integrated analysis of how ATC techniques interact with the broader reliability concerns at both the chip and package levels. Specifically, their work lacks a detailed discussion on how these techniques interact with specific failure mechanisms, such as gate-oxide degradation or short-circuit limitations. Furthermore, they do not explore how ATC strategies can be evaluated through standardized reliability tests.

Therefore, this article aims to enhance existing research by providing a comprehensive perspective on the reliability of power semiconductors. To achieve this, the article includes a detailed overview of power semiconductor technologies, specifically Si-IGBT and SiC-MOSFET, along with their evolution and impact on reliability. Additionally, it offers a market-oriented analysis of standard semiconductor packages,

enabling readers to relate reliability considerations to available industry solutions. The contributions of this paper are:

- *Comprehensive analysis of power semiconductor reliability:* This article provides a thorough examination of reliability in Silicon (Si) and Silicon Carbide (SiC) devices, including an overview of the evolution of these technologies. It highlights critical reliability aspects, detailing where they have the most significant impact within the system.
- *Classification of Si-IGBT and SiC-MOSFET packages:* A detailed overview of standard packages currently available in the market is presented, helping readers gain insight into the current technological landscape and manufacturer solutions.
- *In-depth analysis of ATC techniques:* This article systematically examines various ATC strategies, emphasizing their reliability implications. Particular focus is given to ATC strategies based on gate voltage variation, offering deeper insights into their application and impact on system reliability.

The diagram in Fig. 1 offers an overview of the article's structure. Section II introduces the foundational concepts of reliability in power semiconductors, along with descriptions of standard reliability tests. Section III addresses chip-level reliability by focusing on Si-IGBT and SiC-MOSFET technologies, highlighting the main chip-level failures and their implications. Section IV shifts attention to package-level reliability, discussing common package failures, essential package components, and a classification of standard power modules currently available in the market. Finally, Section V explores active thermal control strategies, outlining the main ATC techniques along with their advantages and disadvantages, as well as their impact on improving reliability.

II. POWER SEMICONDUCTOR RELIABILITY

Reliability is the ability of a system or component to perform its required functions under stated conditions for a specified

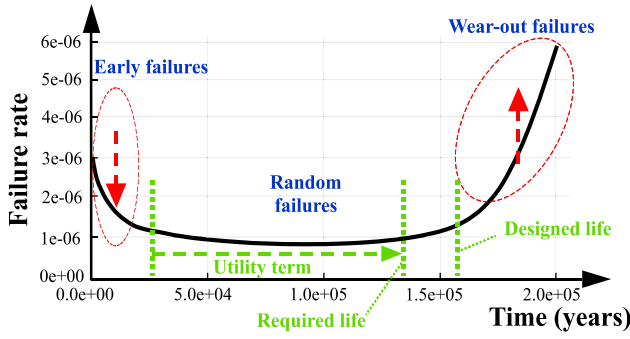


FIGURE 2. Bathtub curve of the failure rate for semiconductor devices with respect to time. Adapted from [19], [26].

period of time [18]. Various metrics such as the frequency of failure, probability of failure, or item availability are used to measure reliability [15], [19]. The electric transport sector for example, including electric vehicles (EV) and trains, is subject to high reliability constraints since any failures could lead to disastrous accidents [20]. In the energy sector, power converters are critical components that ensure that the grid codes are respected, particularly during grid faults or anomalies, such as Low-Voltage Ride-Through (LVRT) events [21]. Therefore, converters need to meet high reliability standards. The failure rate λ (also Failures In Time (FIT)) is described as one failure in 10^9 hours. Here, it is important to note that this is a statistical parameter, meaning that based on the failure rate, it is not feasible to forecast when a failure is happening but rather the likelihood of a failure happening in a defined time interval [22]. Typically, for equipment that demands high reliability, a semiconductor device failure rate of 10 to 100 FIT is required [18].

The curve of failure rate for semiconductor devices can be divided into three stages according to time: Early Failures (or infant mortality phase), Random Failures (or normal life) and Wear-out Failures. Due to the peculiar shape of the failure rate during time, this curve is called the bathtub curve, as shown in Fig. 2.

The failure rate of power semiconductor devices is generally higher at the beginning and end of their useful life [19]. This is due to the fact that defective devices often have a higher probability of failure at the early stage of their life, which is caused by microscopic defects in the manufacturing process such as wafer defects [19]. To reduce early failures, improving quality control and removing defective parts after outgoing tests is important [23]. For example, screening processes on production lines aim to identify and remove faulty devices before they are released to the market [24]. Subsequently, the failure rate falls exponentially and remains constant over a long period of time, this stage is called Random Failures stage. Here, most of the pre-damaged devices have already failed, but wear-out effects are not yet a dominating factor [22]. The failures in this zone are mainly linked to cosmic rays, which can cause ionization events within the

semiconductor material. In high-altitude applications, such as aviation, aerospace, or satellite operations, cosmic rays are more intense and can interfere with the device's ability to block electric fields. This interference can lead to transient or permanent breakdowns, potentially causing system malfunctions or catastrophic failure [25].

The third phase is the Wear-out Phase, where the product fails due to age and wear mechanisms, and the failure rate rapidly increases. Between the normal life and the Wear-out Phase is the lifetime. In general, manufacturers intend to design a system that does not experience wear-out effects during its manufacturer-defined lifetime [15].

To quantitatively describe the behaviour of failure rates over time, the Weibull distribution is widely adopted [24], [26], [27]. The probability density function is defined as:

$$f(t) = \frac{b}{T} \left(\frac{t}{T} \right)^{b-1} \cdot \exp \left[- \left(\frac{t}{T} \right)^b \right] \quad (1)$$

where b is the shape factor, t is time, and T is the scale factor. From this, the survival probability $R(t)$ and the instantaneous failure rate $\lambda(t)$ can be derived as:

$$R(t) = \exp \left[- \left(\frac{t}{T} \right)^b \right], \quad \lambda(t) = \frac{f(t)}{R(t)} = \frac{b}{T} \left(\frac{t}{T} \right)^{b-1} \quad (2)$$

For example, in a hypothetical power electronic system presented in [26], with Weibull parameters derived from field data, the survival probability after 156,000 h of operation is approximately 87.8%. If a system operates continuously 24/7 for 20 years (175,000 h), the survival probability drops slightly to around 85%. In contrast, a system operating in industrial conditions (8 h/day, 5 days/week) would accumulate 41,000 h over 20 years, yielding a survival rate as high as 94%.

When manufacturers provide the bathtub curve or failure rates, they assume that the device operates within the specified limits [12], [28]. If these limits are exceeded, the manufacturer can no longer guarantee the predicted failure rate [29], [30]. These operational limits are defined by the Safe Operating Area (SOA), which ensures the device operates safely within its designed boundaries [12]. Operating outside the SOA can lead to accelerated degradation or even immediate failure of the device [31]. An example of a SOA curve for a commercial SiC-MOSFET module is shown in Fig. 3, illustrating the typical operational boundaries and switching conditions defined by manufacturers [32]. The figure highlights two limits: one for the bare chip and another for the full module. The difference mainly comes from the effect of parasitic inductances, which are explored further in Section IV.

The SOA of a semiconductor is the range of voltage and current conditions where it can operate without risk of destruction. Multiple curves may be provided for different pulse widths, and separate SOA graphs may exist for different temperatures [31]. SOA curves typically represent voltage on the

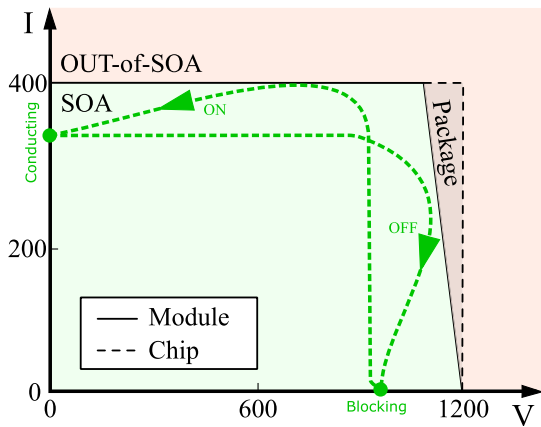


FIGURE 3. SOA of FF4MR12W2M1H_B70 EasyDUAL module based on datasheet values [32]. The ON and OFF transitions are shown along with package limitations.

X-axis and current on the Y-axis. These curves provide different regions based on operating modes and durations, such as continuous, pulsed, or short-circuit operations. Manufacturers provide different types of SOA curves depending on the device's intended usage scenarios. Some of the most commonly used SOA are the Forward-Bias SOA (FBSOA), Short-Circuit SOA (SCSOA), and Reverse-Bias SOA (RBSOA). Among them, the most frequently used type is the Reverse-Bias SOA, also known as turn-off SOA, which is typically provided by manufacturers [32], [33].

It is important to consider that power modules may encounter operating conditions where certain limits are exceeded, resulting in overloads. When a semiconductor device is subjected to an overload event, a clear understanding of these limits is crucial. For instance, the voltage axis (X-axis) has a very narrow margin for exceeding its limits, as a voltage overshoot of only 40–50% beyond the rated value can lead to immediate breakdown failure, even in new modules [34], [35]. On the other hand, the current axis (Y-axis) provides more flexibility. When it comes to exceeding the current limits, it is important to note that there are different limits, such as those imposed by the terminals [33]. However, the primary concern is the junction temperature (T_j) of the device, as surpassing the current limits can lead to long-term degradation and potential failure if the device is not properly managed. A more detailed analysis of current-related limits is provided in Section IV-A5.

The T_j is primarily influenced by the semiconductor's power dissipation, which depends on its losses, as well as the efficiency of heat transfer from the junction to the ambient. This heat transfer process is determined by the thermal conductivity of the materials within the package, including the junction-to-case interface, the case-to-heat-sink connection, and the heat sink itself, along with ambient conditions. The thermal pathway is quantified by the thermal resistance (R_{th}), which can be decomposed into different components corresponding to each interface within the package. Each element in the package contributes to the overall thermal resistance,

ultimately affecting the device's ability to manage temperature and ensuring its reliability. A more detailed description of the package elements and their thermal influence is provided in Section IV.

A. STANDARD RELIABILITY TESTS

To validate their designs and minimise the chances of failures during operation, manufacturers conduct standard reliability tests [23]. These tests are specifically designed to accelerate the failure mechanisms that are typically encountered in real-world applications [24]. By subjecting the Device Under Test (DUT) to extreme operating conditions, these tests ensure that they meet the reliability standards for their intended use.

Reliability testing involves applying controlled stress profiles, including thermal cycling, high humidity, mechanical vibration, and rapid temperature changes. During these tests, key parameters are monitored, such as thermal impedance, leakage currents, on-state voltage, and shifts in threshold voltages, to evaluate the device's behaviour under stress. The main goals of these tests are to identify critical failure mechanisms and predict the operational lifetime of the device. They also aim to verify stability and ensure compliance with quality standards, while establishing operational constraints and performance limits for real-world applications.

These stress tests are especially relevant for applications such as electric vehicles (EVs), railway traction, and aerospace, where frequent power cycling, wide temperature swings, and intense mechanical vibration make thermal cycling and vibration tests essential [20], [36], [37]. In wind and PV energy systems, converters are typically installed outdoors and face wide daily temperature gradients due to ambient variation between day and night, increasing the relevance of thermal fatigue and humidity-related degradation [38]. Therefore, tests such as High Temperature Storage (HTS), High Humidity High Temperature Reverse Bias (H3TRB), and Temperature Cycling are crucial to ensure long-term reliability in these sectors [39], [40]. Although stress profiles may vary, many of these applications share test requirements due to similar degradation factors.

Many organizations have established widely recognized reliability standards to guide evaluations of electronic components. For example, the Joint Electron Device Engineering Council (JEDEC) [41] and the International Electrotechnical Commission (IEC) [42] provide protocols for testing components under harsh conditions. In the automotive industry, guidelines such as AQG 324 from the European Center for Power Electronics (ECPE) [43] and the Automotive Electronics Council (AEC) [44] are commonly adopted to address the challenges faced by power semiconductor modules.

In the following sections, we will discuss specific test categories, including environmental tests, lifetime tests, and various test bench configurations that will be shown in order to perform power cycling test. A summary of these reliability tests, their objectives, and key conditions is provided in Table 1.

TABLE 1. Summary of Standard Reliability Tests for Power Semiconductor Modules

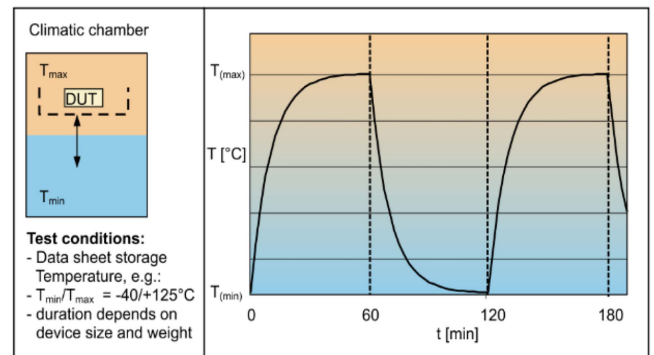
Test Type	Purpose / Description	Typical Conditions / Failure Criteria
Thermal Cycling (TC)	Assess mechanical stress due to passive temperature changes.	-40 °C to 125 °C until thermal resistance (R_{th}) increases by more than 20%.
High Temperature Storage (HTS)	Assess degradation due to prolonged exposure to high ambient temperatures.	>125 °C for 1000 h. DUT must remain fully functional and undamaged.
Low Temperature Storage (LTS)	Evaluate robustness to cold storage or transport.	<-40 °C for 1000 h. DUT must remain fully functional and undamaged.
Vibration Test	Evaluate mechanical integrity under vibration stress, especially for automotive applications.	10–1000 Hz, 5 g acceleration. DUT must remain fully functional and undamaged.
Mechanical Shock	Assess mechanical resistance to sudden acceleration/shocks.	Conducted without electrical operation. DUT must remain fully functional.
High Temperature Reverse Bias (HTRB)	Stress insulation and passivation layers under maximum reverse bias.	80–100% V_{BR} , near $T_{j(max)}$, for 1000 h. Monitor for leakage current increase.
High Temperature Gate Bias (HTGB)	Evaluate gate oxide degradation under prolonged gate bias.	$\pm 100\%$ $V_{GE(max)}$, near $T_{j(max)}$, 1000 h. Monitor V_{th} shift.
High Humidity High Temp. Reverse Bias (H3TRB)	Test moisture-induced degradation pathways.	85 °C, 85% RH, for 1000 h under reverse bias. Monitor for leakage current increase.
Power cycling (PC)	Evaluate thermal fatigue due to active heating/cooling cycles.	Fast: $\Delta T_j > 100$ °C (wire-bond stress); Slow: $\Delta T_j < 80$ °C (solder fatigue).

1) ENVIRONMENTAL TESTS

Environmental tests are used to verify the suitability of power electronics modules under extreme thermal and mechanical conditions. Physical analyses, verification of electrical and mechanical parameters, and insulation properties are used for the validation.

Thermal Cycling: In temperature cycling tests the temperature of a power module is changed by variation of the ambient temperature (TST: Thermal Shock Test) or the temperature of the case (TC: Thermal Cycle Test) without electrical stressing. In this test, components are periodically moved up and down between a cooling chamber and a heating chamber with the help of a lift cage. The TC test is applied mainly to evaluate the lifetime of the solder joints and to evaluate the resistance to the sudden changes in temperature the device can experience during storage, transportation or in use. This test validates the resistance to mechanical stress from passive temperature changes in the case itself and between layers with different thermal expansion coefficients. In particular, large solder areas between direct bond copper and copper base plate are under stress due to these thermal expansions and contractions. The test duration must be sufficient to ensure that the DUT is fully stabilized at both the low and high temperature extremes. The device should remain at each temperature level for more than 15 minutes, and a typical test setup uses 1–2 temperature cycles per hour [43]. The standard temperature range for testing power modules is from -40 to 125. Modern devices, especially those using sintered dies and no baseplate, can be tested for over 1500 cycles without failure [12]. Fig. 4 shows the test set-up and the evolution of the temperature over time. A thermal resistance increase of 20% compared to the initial value before the test is considered a failure.

High Temperature Storage: The purpose of this test is to test or determine the effect of storage at increased temperature by exposing the case to high thermal stress, with an


FIGURE 4. Temperature cycling test set-up and evolution of temperature over time [12].

ambient temperature above 125 and a total duration of 1000 hours [12], [43], [45]. The DUTs must be fully functional after the test, meaning they should not show any signs of damage at the end.

Low Temperature Storage: The purpose of this test is to test or determine the effect of aging or transport at a very low temperature on the power modules, typically with temperatures around -40 and a duration of 1000 hours [12], [43], [45]. As well as the high-temperature storage test, at the end of the test, the cases must not show any signs of damage, and they must be fully functional.

Vibration Test: The purpose of the test is to show the fundamental suitability of the mechanical structure for use in automotive power electronics converter units. Typical weak points in the mechanical structure are the mechanical aging of spring contacts, the stability of solder contacts in case of vibrating masses and cracks in cases of structural parts [43]. This test is performed in a frequency range between 10 Hz and at least 1000 Hz using an acceleration force of 5 g. Some

limitations regarding the minimum frequency may emerge depending on the test equipment. At the end of the tests, the device must be fully functional and all parameters must meet the specifications.

Mechanical Shock: The mechanical shock test aims to verify the DUT's shock resistance and the solder joints' ability to withstand external stress during shock. It also assesses the power device's durability to mechanical shock and potential failure patterns like cracks or detachment [43]. The tests must be conducted without electrical operation, and the device should be fully functional and meet all specifications.

2) LIFETIME TESTS

Unlike environmental tests in which components are passively tested, in lifetime tests the DUTs are tested under different conditions of use. The lifetime tests are intended to activate the typical degradation mechanisms of power electronics modules.

High Temperature Reverse Bias: The High Temperature Reverse Bias test (HTRB) evaluates long-term stability under high drain-source/collector-emitter bias in order to determine weak points in the passivation layer structure and at the chip edge sealing over time [43]. During this test, the DUTs are stressed by applying between 80% and 100% of the maximum reverse breakdown voltage (V_{BR}), with the gate shorted to the emitter, at an ambient temperature close to the maximum rated junction temperature, for a total duration of 1000 hours. The leakage current of a stable device should remain relatively constant, only increasing slightly over the testing period. A failure occurs when leakage rises to a level that triggers thermal runaway [31]. In order to avoid unexpected breakdown, voltage stress is controlled by limiting the applied voltage preferably to 80% of V_{BR} , since testing closer to 100% increases the risk of cosmic ray-induced failures without providing additional reliability information [43]. Furthermore, leakage current is continuously monitored, and in advanced test setups, a soft bias approach with gradual voltage ramp-up can be used to minimize initial overstress [46].

High Temperature Gate Bias: The High Temperature Gate Bias (HTGB) test is specifically utilised for semiconductor devices that feature a gate connection. Its primary purpose is to analyse the degradation of these devices over time, focusing on the combined effects of electrical and thermal stress. The test is designed to detect for drift caused by random oxide defects and ionic oxide contamination. Voltage is applied between the gate and emitter with the collector shorted to the emitter and the variation of the threshold value (V_{th}) is monitored device. During this test, the DUTs are stressed by applying 100% of the maximum and minimum rated gate-emitter voltage, an ambient temperature close to their maximum rated junction temperature and a total duration of 1000 hours.

High Humidity High Temperature Reverse Bias: The presence of moisture under bias in all electronic devices has been a historical issue. Semiconductors such as conventional

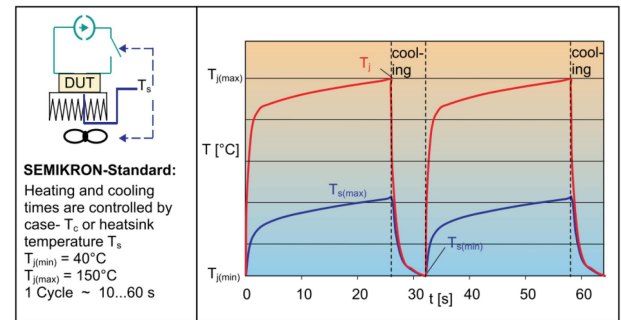


FIGURE 5. Power cycling test set-up and evolution of temperature over time [12].

Si power devices introduce the added complexity of higher voltages present at the exposed chip surfaces as well as materials and active regions that are sensitive to humidity-induced degradation. As most module designs are not hermetically sealed and semiconductor chips and bonding wires are embedded in silicone gel, which is permeable to humidity, moisture reaches the passivation layer over time [48]. The high humidity, high temperature reverse bias (H3TRB) test also known as temperature and humidity bias (THB) is a test that determines weak points in the overall module structure, including the power semiconductor itself [43]. This test is made in an environmental chamber set to 85% relative humidity (RH) and 85 ambient temperature where the DUTs are stressed for 1000 hours. The variation of the drain-source leakage current is monitored during the test.

Power cycling: The temperature of a power semiconductor varies with operating conditions. During power cycling tests (PC tests), components are heated by semiconductor losses and then cooled, termed “Active” cycling tests, while temperature cycling is called “Passive” [12]. Devices are mounted on heat sinks, and when the maximum target temperature is reached, the load current is switched off to allow cooling. This creates significant internal temperature gradients during repeated cycles [25]. Fig. 5 shows the evolution of temperature over time in a power cycling test.

Fast power cycling (tens of seconds) and larger temperature swings ($\Delta T > 100^\circ\text{C}$) can cause wire-bond failures, while slower cycling (minutes) and smaller swings ($\Delta T < 80^\circ\text{C}$) induce solder fatigue [49]. Results are shown as curves of maximum cycles versus temperature differences, using high ΔT values for rapid testing, which are then extrapolated to real-world conditions.

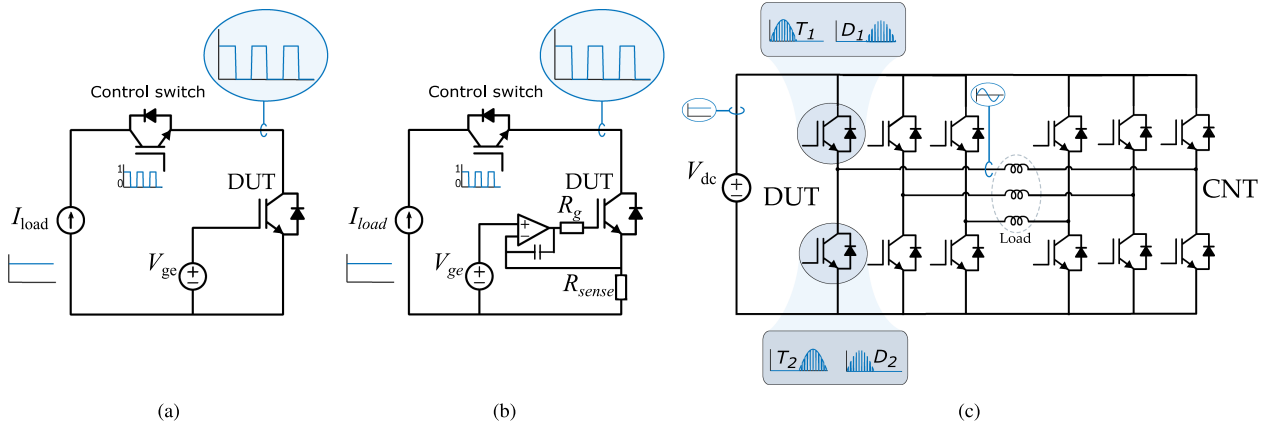
According to ECPE Guideline [43], power cycling splits into two tests: PCsec and PCmin. PCsec uses power pulses below 5 seconds, impacting chip-near interconnections, while PCmin applies pulses over 15 seconds, affecting both remote and near interconnections.

3) POWER CYCLING TEST BENCH CONFIGURATION

Power cycling tests are crucial for evaluating the reliability of power semiconductor devices under repeated thermal stress.

TABLE 2. Comparison Between Different Power Cycling Test Configurations

Aspect	DC (Conventional)	DC (Saturation)	AC (Back-to-back)
Switching losses included	Only conduction losses	Only conduction losses	Both conduction and switching losses
Realistic electrical conditions	No	No	Yes
Setup complexity	Low	Medium	High (requires inverter control)
Current requirement	High	Lower than conventional DC	Low
Cost	Low setup cost, high energy cost	Medium cost	High setup cost, low energy cost


FIGURE 6. Power cycling circuit configurations [47]: (a) Conventional DC configuration; (b) Saturation-mode DC configuration with gate-controlled losses; (c) AC configuration using a back-to-back inverter setup.

These tests are designed to accelerate the degradation processes caused by temperature fluctuations, particularly those affecting bond wires, die-attach layers, and solder joints [47]. The primary goal is to subject the device to cyclic thermal loading by repeatedly heating it through electrical operation until it reaches a target junction temperature, followed by a cooling period, which simulates real-world operating conditions [12]. There are two main test configurations used: DC-based and AC-based power cycling circuits. Table 2 provides a summary of the main features of each configuration, while Fig. 6 represents the circuit configurations.

Conventional DC Circuit: This configuration (Fig. 6(a)) is the most common approach [50], [51]. The junction temperature increases due to conduction losses during the ON phase and decreases during the OFF phase, regulated by external cooling systems. Although this configuration is cost-effective and simple, it only accounts for conduction losses, meaning results are not fully representative of real-world conditions. Furthermore, achieving high temperatures requires injecting large currents, which risks overloading bond wires [47].

Saturation mode DC Circuit: The saturation mode DC configuration (Fig. 6(b)) operates the DUT in its active region to induce power losses without requiring high load currents [47]. In this approach, the collector current is modulated through the gate-emitter voltage to reach the target junction temperature. This configuration enables the use of a less demanding test bench, as it achieves comparable thermal stress levels with lower current amplitudes. However, it remains a complex setup and is not compatible with modules that integrate

internal gate drivers. Moreover, like the conventional DC method, it still does not replicate realistic converter operating conditions.

AC circuit: In contrast, AC configurations (Fig. 6(c)) replicate realistic operating conditions by including both conduction and switching losses. This configuration typically involves a back-to-back inverter setup with inductive loads, minimising energy input and enabling long-duration testing [47], [52]. The load inverter of the back-to-back configuration, also known as the control unit (CNT), is designed to effectively manage power flow and thermal stresses during testing. Employing a higher rated-power than the DUT reduces the impact of thermal stresses experienced in the PC tests. AC test benches are more complex and costly to implement, requiring sophisticated controllers and higher-rated power devices to withstand thermal stresses during testing. However, testing under realistic electrical conditions is essential for applications requiring high reliability, making the AC configuration the preferred choice for power cycling tests [52], [53].

B. LIFETIME CURVES

The connections inside power modules age over time due to changes in temperature. The materials in the connections become fatigued due to thermal stress resulting from the temperature gradient and different coefficient of thermal expansion (CTE) of the connected materials [54]. The module's lifetime, or the number of possible temperature cycles it can experience, decreases as the temperature difference (ΔT)

TABLE 3. Parameters and Limits for Power Cycling Lifetime model [12], [59]

Parameter	Symbol	Limit	Coefficient	Unit
Temperature swing	ΔT_j	45–150	$\beta_1 = -4.416$	K
Min. junction temp	$T_{j,min}$	20–120	$\beta_2 = 1285$	°C
Pulse duration	t_{on}	1–15	$\beta_3 = -0.463$	s
Current per bond	I_B	3–23	$\beta_4 = -0.716$	A
Voltage class/100	V_C	6–33	$\beta_5 = -0.761$	V
Bond wire diameter	D	75–500	$\beta_6 = -0.5$	μm

increases [55], [56]. Power cycling is a process that raises and lowers the temperature of the chip-junction at short intervals in a matter of seconds. This process applies stress to the bond wires on the silicon chips and the soldered joints below them.

The power-cycling capability of power semiconductor devices depends on multiple thermal and electrical factors. One of the most widely used models for estimating the number of cycles to failure (N_f) is derived from the Coffin–Manson law, originally developed for fatigue in solid materials [54], [57], [58]. In the 1990s, this approach was extended through the LESIT project to include the effect of the average junction temperature ($T_{j(av)}$) by incorporating an Arrhenius-type acceleration factor [50]. This leads to a semi-empirical model known as the LESIT equation:

$$N_f = A \cdot (\Delta T_j)^\alpha \cdot e^{\left(\frac{E_a}{k_b \cdot T_{j(av)}}\right)} \quad (3)$$

In this expression, A and α are empirical fitting parameters obtained from test data, E_a is the activation energy, and k_b is the Boltzmann constant [12]. This model is commonly used in industry due to its simplicity and its ability to capture the two most dominant thermal factors impacting lifetime: thermal cycling amplitude and mean junction temperature [50].

However, in modern applications, the influence of other parameters, such as heating time (t_{on}), current per bond wire (I_B), blocking voltage (V_C), and bond wire diameter (D) can be also considered [57]. To improve lifetime estimation accuracy, authors in [59] proposed an extended version of the LESIT model known as the CIPS08 model, which incorporates these additional factors:

$$N_f = k_b \cdot (\Delta T_j)^{\beta_1} \cdot e^{\left(\frac{\beta_2}{T_{j,min} + 273}\right)} \cdot t_{on}^{\beta_3} \cdot I_B^{\beta_4} \cdot V_C^{\beta_5} \cdot D^{\beta_6} \quad (4)$$

This equation is built upon extensive power cycling test data, with the coefficients β_i determined through regression analysis. A summary of typical values and their limits can be found in Table 3. By considering electrical and packaging effects explicitly, this model provides more accurate lifetime predictions for specific module technologies and operation profiles [59].

While the simplified LESIT model (3) is often used in early-stage design, especially when limited data is available, the extended model (4) offers higher accuracy when detailed test parameters are known.

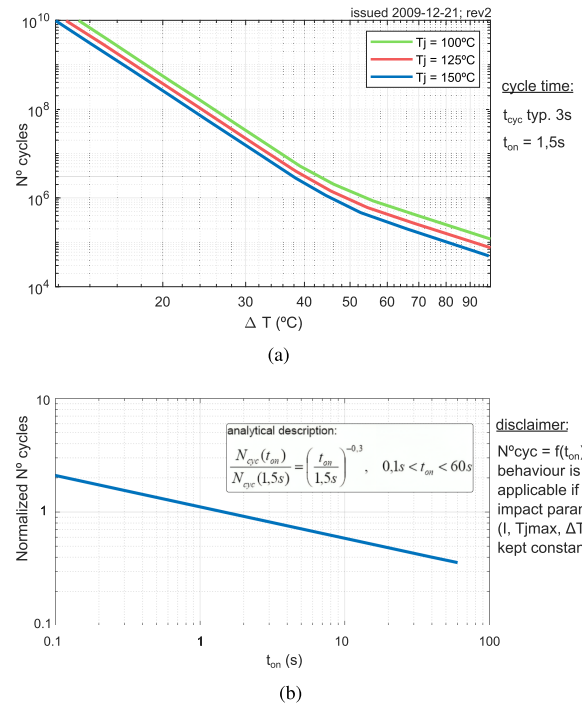


FIGURE 7. Power cycling curves provided by manufacturer (based on [60]): (a) Typical lifetime curves, and (b) normalization of the cycle as a function of its duration.

In addition to these models, manufacturers often provide empirical lifetime curves based on extensive internal testing [12], [23], [60]. Fig. 7(a) illustrates a typical power cycling capability chart for industrial IGBT4 modules from Infineon, showing the expected number of cycles as a function of the junction temperature swing (ΔT_j) for different maximum junction temperature ($T_{j,max}$) levels [60]. These curves offer designers a practical reference to assess device reliability under target thermal conditions and complement the use of analytical models. Moreover, normalization methods such as the one shown in Fig. 7(b) allow adapting lifetime curves to specific application conditions in terms of pulse duration. Here, t_{on} refers to the duration during which the device generates power losses, resulting in a rise in junction temperature T_j . For a constant temperature swing ΔT_j , a shorter t_{on} induces faster thermal transitions, which in turn increases thermo-mechanical stress. Conversely, longer t_{on} periods allow more gradual heating and cooling, thereby reducing the stress. This behaviour is valid only if other influencing parameters (e.g., current, $T_{j,max}$, and ΔT_j) remain constant.

C. RELIABILITY HANDBOOKS AND STATISTICAL MODELS

Along with previously mentioned experimental testing and physics-of-failure (PoF) models, which use accelerated stress tests, such as power cycling, to characterize specific degradation mechanisms, another common method for estimating reliability is based on statistical handbooks. Notable examples include MIL-HDBK-217, FIDES, Telcordia SR-332 and SN29500 [17], [22], [27].

These handbooks rely on large databases of field failure data and provide base failure rates combined with empirical acceleration factors to estimate metrics such as the Failure in Time (FIT) rate or Mean Time Between Failures (MTBF). In contrast to the time-dependent failure rate $\lambda(t)$ used in Weibull-based PoF models, handbook-based methods assume a constant component-level failure rate λ_i , typically defined as [61]:

$$\lambda_i = \lambda_b \cdot \pi_T \cdot \pi_A \cdot \pi_S \cdot \pi_Q \cdot \pi_E \quad (5)$$

where λ_b is the base failure rate, π_T accounts for operating temperature, π_A for application type (e.g., power class), π_S for voltage stress, π_Q for part quality, and π_E for environmental conditions.

This approach is particularly convenient during the early design stages due to its simplicity and ability to support a parts count method across a bill of materials (BOM) [22]. However, it does not consider degradation phenomena such as thermo-mechanical fatigue or gate oxide wear-out. As a result, handbook-based models neglect the wear-out phase typically captured by PoF-based approaches, and instead assume a constant failure rate throughout the useful life phase of the bathtub curve [27].

MIL-HDBK-217, for instance, was officially withdrawn by the U.S. Army for reliability prediction purposes in 1995 due to its limitations [27]. Newer standards, such as IEC 61709, attempt to address these drawbacks by incorporating mission profiles and temperature cycling factors [17]. Additionally, Monte Carlo-based reliability assessments are gaining relevance as they allow probabilistic treatment of uncertainties in lifetime models, operating conditions, and manufacturing tolerances [27].

III. CHIP-LEVEL RELIABILITY

Traditionally, the field of power electronics has relied on silicon-based devices, such as MOSFETs and IGBTs. However, in recent years, improvements in manufacturing processes have opened up the way for wide bandgap (WBG) power semiconductor devices [24]. These devices are based on materials like silicon carbide (SiC) and gallium nitride (GaN), which can operate at very high working frequencies, voltages, and temperatures. As their construction issues are resolved, and as they reach the current-voltage ranges of silicon-based semiconductors, the WBG devices are going to replace the silicon-based devices in various applications [62]. However, using these devices poses a series of challenges that need to be addressed and solved to fully exploit their potential. Si-based devices are a very mature technology, and so they still dominate the market of power devices due to their reduced cost and ruggedness.

A. SI-IGBT

The Insulated Gate Bipolar Transistor (IGBT) was introduced in the 1980 s by General Electric as a hybrid device combining the high-input impedance of MOSFETs with the low

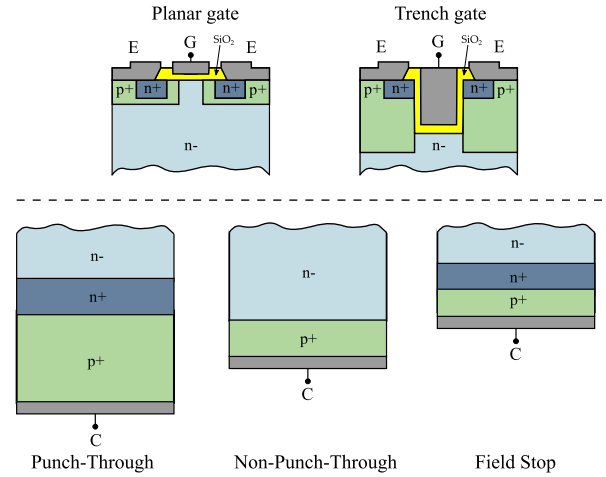


FIGURE 8. Classification of IGBT based on their internal structure.

conduction losses of BJTs [63]. This is achieved by adding a p^+ collector layer to a vertical MOSFET structure, enabling bipolar conduction while maintaining voltage-controlled operation.

IGBTs are widely used due to their ability to handle high voltages and their low on-state voltage drop. Si-IGBTs have become the preferred option above 1000 V, particularly in motor drives, renewable energy systems, and inverters. They can be classified by gate structure (planar or trench) and vertical structure (Punch-Through (PT), Non-Punch-Through (NPT), and Field-Stop (FS)). Fig. 8 shows this classification.

Fig. 9 summarizes the evolution of IGBT technology, showing changes in structure, wafer size, and operating temperature [20], [63], [64].

The first IGBT generation, the PT structure, used an epitaxial substrate with planar gates and a buffer layer to manage the electric field. These devices operated up to 1700 V but required large silicon volumes, increasing cost. Reliability challenges included strong temperature dependence of electrical parameters and limited short-circuit capability, complicating parallel operation and increasing failure risk under overload [25], [63].

To overcome these issues, the NPT structure was developed by Siemens (now Infineon) [25]. It eliminated the epitaxial layer, reducing cost and enabling voltage scalability up to 4500 V. NPT devices also offered improved parameter uniformity, positive temperature coefficients, and better short-circuit robustness due to the use of Float Zone (FZ) substrates [20], [63], [65]. However, they showed increased conduction and switching losses compared to PT devices.

The FS structure was introduced as an improvement over both PT and NPT IGBTs. ABB patented the Soft-Punch-Through solution in 2000, and other manufacturers have used the same concept by using other structure names like Low-Punch-Through or Thin-Punch-Through. They all have in common a n^- , n^+ and p^+ reduction in wafer thickness with respect to traditional solutions. In 2002, Infineon created the

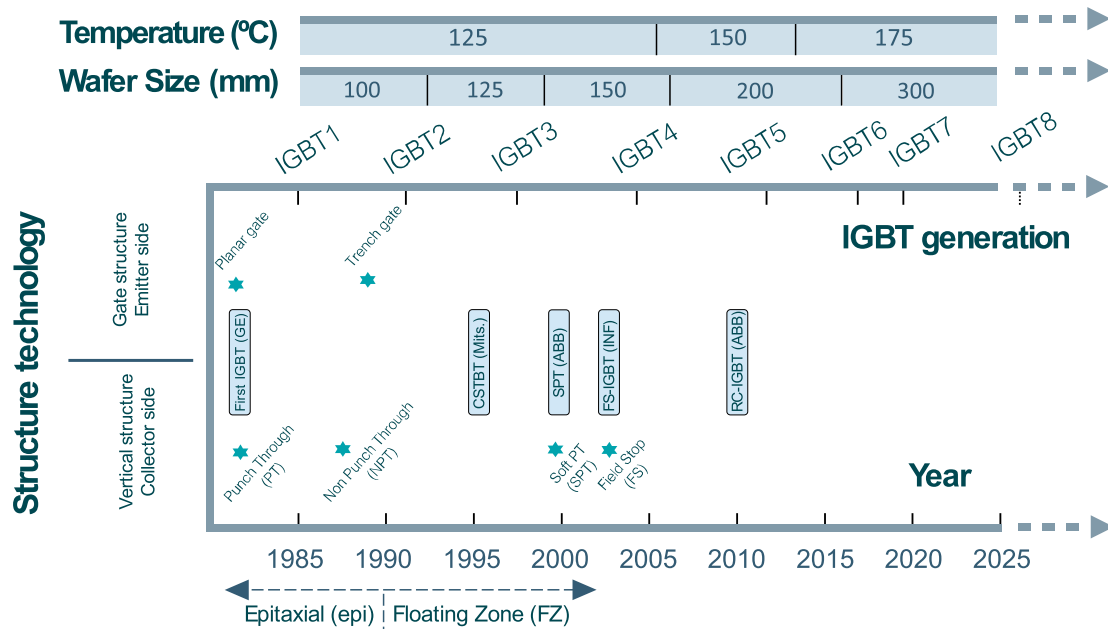


FIGURE 9. Evolution of IGBT technology. Adapted from [20], [63], [64].

Trench-Field-Stop or TrenchSTOP structure, which differs in gate structure (trench instead of planar). Depending on the manufacturer, it may also be referred to by other names such as Soft-Punch-Through (SPT) and Light-Punch-Through (LPT) from ABB and Mitsubishi respectively.

The semiconductor industry has also transitioned from 100 mm to 300 mm wafers, improving manufacturing yield and cost-effectiveness [62], [66]. Simultaneously, die area reduction has minimized defect probability, while packaging advancements have enhanced maximum junction temperature and overall reliability, as further discussed in Section IV.

Recent developments combine Enhanced Trench structures with FS designs. For instance, Infineon's Micro Pattern Trench (MPT) increases channel width to reduce on-state voltage drop [20], while Mitsubishi's CSTBT integrates Carrier Stored layers to enhance short-circuit tolerance and expand the safe operating area (SOA) [67]. Other innovations include Reverse Blocking (RB) and Reverse Conducting (RC) features. The latter, combined with an anti-parallel FreeWheeling Diode (FWD), is crucial in applications like EVs. Fig. 11 illustrates the RC-IGBT structure with integrated n -shorts for reverse conduction.

B. SiC-DEVICES

While this review ultimately focuses on the reliability challenges of SiC-MOSFETs, it is important to understand the broader technological evolution of SiC devices. Early developments, including diodes and JFETs, laid the foundation for current SiC-MOSFETs and its associated reliability concerns. Fig. 10 provides a summary of this evolution.

SiC is a wide bandgap material that offers higher critical electric field, better thermal conductivity, and lower intrinsic carrier concentration than silicon [69]. Table 4 compares the

TABLE 4. Comparison Between the Electrical Properties of Si and WBG Materials [69], [70], [71]

Material	W_g (eV)	E_{crit} (MV/cm)	μ_n (cm ² /Vs)	ϵ_r (-)	n_i (cm ⁻³)
Si	1.12	0.29	1350	11.9	1×10^{10}
3C-SiC	2.35	1.5	900	9.7	1.5×10^{-1}
4H-SiC	3.28	2.2	800	9.6	8×10^{-9}
6H-SiC	2.96	3.2	370	9.6	5×10^{-9}
GaN	3.4	2	1700	9	2×10^{-10}
Diamond	5.5	20	2200	5.7	1×10^{-20}

key electrical properties of Si, GaN, and SiC-based materials [69], [70], [71].

Among these, SiC stands out for high-voltage applications. Its higher E_{crit} enables lower drift region resistance, while low n_i and high thermal conductivity support higher operating temperatures. GaN, in contrast, is more limited in breakdown voltage due to its lateral structure [70].

The first SiC power device was a Schottky Barrier Diode (SBD), introduced by Infineon in 2001. Although it offered advantages in high-frequency operation, early versions suffered from leakage current degradation due to surface defects and localized electric field concentrations [72], [73]. To address this, the Merged-PiN Schottky (MPS) diode structure added p-wells to redistribute the electric field and improve surge current capability [73].

In 2008, SemiSouth introduced SiC-JFETs, offering fast switching and low conduction losses. However, their normally-on behavior posed safety concerns during converter startup. Cascode configurations helped mitigate this but introduced parasitics and thermal mismatch issues. Despite better

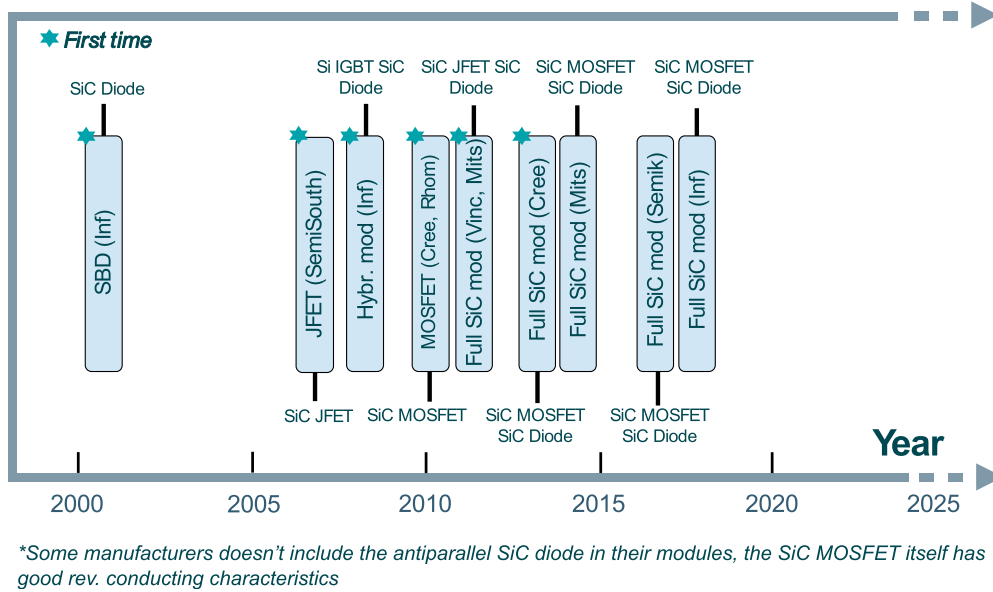


FIGURE 10. Evolution of SiC technology. Adapted from [37].

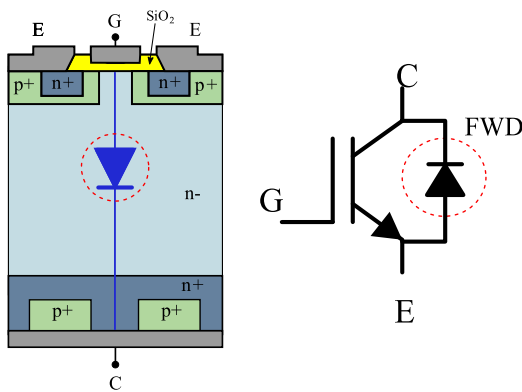


FIGURE 11. Basic structure of an RC-IGBT. Adapted from [12], [68].

reliability due to the absence of a gate oxide, they lost prominence as SiC-MOSFETs matured [74].

The first commercial SiC-MOSFETs were launched between 2010 and 2011 by Rohm and Cree. Initially, their adoption was limited by gate oxide reliability. Compared to Si, the higher electric fields in SiC-MOSFETs increased susceptibility to Time-Dependent Dielectric Breakdown (TDDB) and threshold voltage drift, both of which degrade performance over time [75], [76]. These reliability aspects are further detailed in Section III-C.

Since 2011, SiC power modules using full SiC transistors and diodes have become widespread. Vendors such as Vincotech, Microsemi, Cree, Mitsubishi, Semikron, Fuji, and Infineon have released solutions with improved robustness. Advancements like trench-gate SiC-MOSFETs have reduced defect densities and improved channel conductivity, overcoming early gate oxide concerns and enabling high-frequency and high-temperature operation [77].

Advancements in gate-oxide processing and the introduction of trench-based SiC-MOSFET structures helped overcome these reliability concerns. The trench design reduced defect densities and improved channel conductivity, enhancing both efficiency and durability [77]. In 2013, Cree introduced a full SiC module using SiC-MOSFETs and SiC diodes, marking a key transition in SiC technology. Since then, other device manufacturers, including Mitsubishi, Semikron, Fuji, and Infineon, have also released full SiC modules. Some manufacturers do not include the antiparallel SiC diode in their modules, as the SiC-MOSFET itself has good reverse conducting characteristics.

C. CHIP-LEVEL FAILURES

Multiple mechanisms can result in degradation at the chip level. The degradation of a chip is accelerated by various factors, including electrical and thermal stress experienced during operation, as well as external radiation. All these factors contribute to the chip's accelerated degradation over time. It is worth noting that while Si and SiC power switches have most of these failure modes in common, significant differences in the gate-oxide interface, chip area, and chip thickness under the same voltage class and current rating occur [15]. As a result, some failure modes that are not significant in Si devices do impact SiC reliability.

1) GATE OXIDE

Gate-oxide degradation is a major reliability concern in semiconductor devices, particularly in SiC-MOSFETs, due to their high electric fields and switching speeds. Several mechanisms contribute to this degradation, each affecting device performance in different ways:

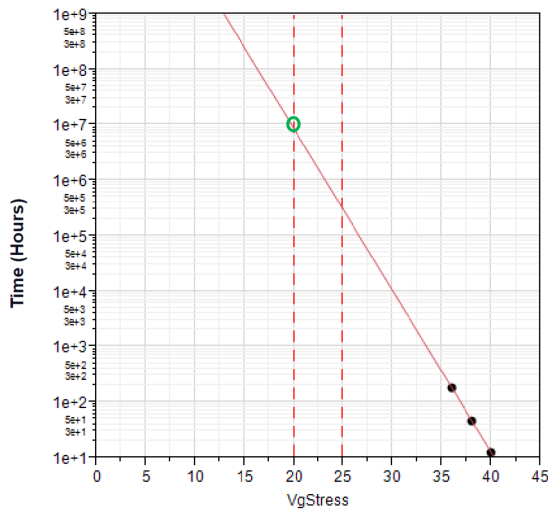


FIGURE 12. TDDB of gate oxide on SiC-MOSFET [78].

Time-Dependent Dielectric Breakdown (TDDB): Caused by long-term electrical stress, TDDB involves the accumulation of trapped charges in the gate oxide. Over time, this leads to the formation of conductive paths, oxide thinning, and eventually gate-to-substrate breakdown [15]. Observable symptoms include increased gate leakage current, decreased drain current and threshold voltage shifts [17].

Hot Carrier Injection (HCI): High electric fields can accelerate carriers (electrons or holes), which then gain sufficient energy to get injected into the oxide. This degrades the oxide quality and accelerates wear-out [17].

Bias Temperature Instability (BTI): BTI is caused by charge trapping at the semiconductor-oxide interface during prolonged biasing at elevated temperatures. This leads to a gradual drift in the threshold voltage, typically manifesting as long-term degradation rather than sudden failure [79].

High Switching Frequency Stress: The repetitive stress associated with fast switching transitions contributes to increased electrical stress on the gate oxide, amplifying the effects of the above mechanisms [15].

To assess these failure modes, several standardized tests are used, including High-Temperature Gate-Bias (HTGB), High-Humidity High-Temperature Gate Bias (H3TGB), High-Temperature Reverse Bias (HTRB), and High-Humidity High-Temperature Bias (H3TB) [12], [15], [43].

Nevertheless, modern SiC technologies demonstrate high gate-oxide reliability. For instance, thermally-grown gate oxides have proven capable of operating reliably under high gate voltages (e.g., 20 V) and elevated temperatures (up to 375 °C) [81]. Fig. 12 shows a lifetime of 10^7 hours under such stress conditions [78]. Further validation of gate-oxide robustness is provided in [82], where four commercial 1200 V SiC-MOSFETs were subjected to high gate voltages over 24 hours. The resulting ramped-voltage measurements showed stable gate leakage current up to 25–30 V, confirming the gate oxide's stability (see Fig. 13).

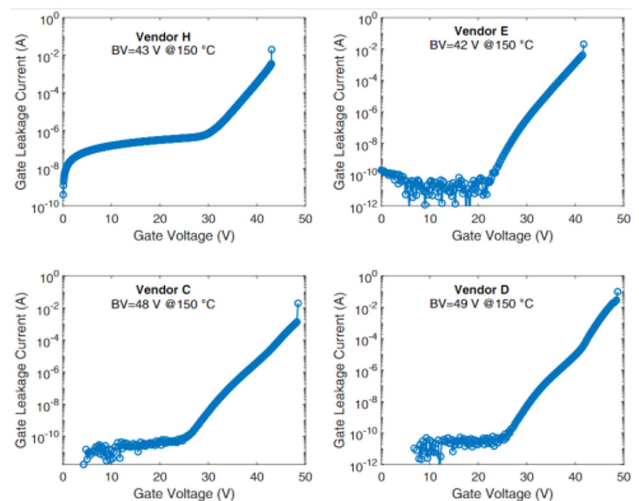


FIGURE 13. Ramped-voltage breakdown measurement results for commercial SiC power MOSFETs [82].

2) GATE THRESHOLD VOLTAGE DRIFT

Threshold voltage drift is a critical issue that is often associated with Bias-Temperature Instability (BTI) and affects both Si and SiC devices. This drift occurs due to defects near the semiconductor-oxide interface, where carriers get trapped and cause a shift in operational parameters. For SiC devices, this instability can be especially problematic because of the material's intrinsic properties and the higher density of interface traps [41]. The drift in threshold voltage increases the on-resistance, and also impacts device performance under varying operational conditions [76]. However, improvements in processing techniques have shown to mitigate these effects, making modern SiC devices increasingly comparable to their Si counterparts in terms of gate stability [15].

3) BODY DIODE RELIABILITY

Another issue associated with MOSFETs is the degradation of the body diode, which is caused by an increase in forward-current. This leads to the formation of a defect known as a stacking defect or stacking faults, which impacts the current flow and results in higher on-resistance and forward-voltage of the body diode [62]. These faults act as barriers to carrier conduction, thus reducing the efficiency of the body diode. The robustness of the body diode can be evaluated through standard tests such as High-Temperature Reverse-Bias (HTRB) and High-Temperature Forward-Bias (HTFB) [41], [62].

4) LATCH-UP

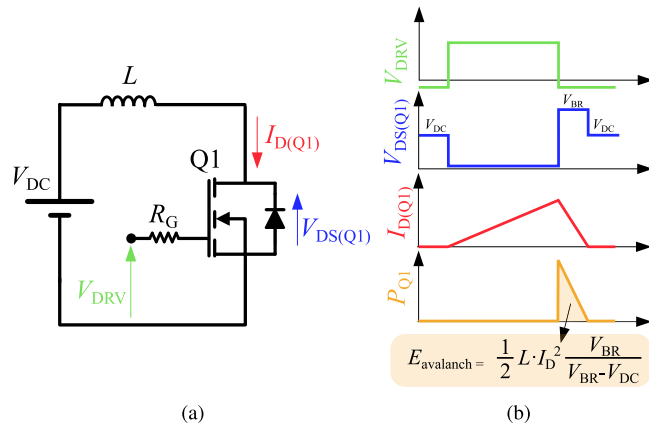
The switch turn-OFF transition under high temperature and high dv/dt can cause latch-up, which is a failure mechanism that can affect both MOSFETs and IGBTs [17], [83]. Latch-up occurs due to the unwanted activation of parasitic bipolar junction transistors (BJTs) in MOSFETs and thyristors in IGBTs. This results in a loss of control over the drain/collector

TABLE 5. Summary of Chip-Level Failure Modes, Associated Stress Tests, Affected Parameters, and Related insights [15], [24], [25], [34], [80]

Failure Modes	Stress Tests	Affected Parameters	Comments
Gate-oxide degradation*	TDDDB	V_g	Critical for SiC devices due to high field stress; detectable via leakage current monitoring. However, screening for oxide defects remains a challenge [15], [25]
	HTGB	V_g, T	
	H3TGB	V_g, T, H	
	HTRB	V, T	
	H3TB	V, T, H	
Gate threshold voltage drift*	DC BTI	V_g, T	Gradual shift in V_{th} leading to $R_{DS(on)}$ increase. Monitoring required over long periods; influenced by processing quality [25].
	AC BTI	V_g, T, f_{sw}	
Body diode degradation**	HTOL (body diode)	T, I_r	Occurs mainly in SiC-MOSFETs due to stacking faults. V_F monitoring provides indirect assessment; fault localization remains difficult [80].
Latch-up	UIS	V	Abrupt and catastrophic failure. UIS testing is essential for qualification, though not predictive for field failures [15].

V_g (gate voltage), T (temperature), H (humidity), I_r (reverse current), V (V_{DS} : SiC-MOSFET, V_{CE} : Si-IGBT), f_{sw} (switching frequency)

*More important for SiC devices than Si devices. **Does not exist for Si-IGBT.


FIGURE 14. (a) UIS test circuit and (b) typical waveforms during the test.

current via the gate voltage. An Unclamped Inductive Switch (UIS) test is performed in order to evaluate the ruggedness and robustness of a power device against latch-up phenomena under avalanche-mode conduction [34]. Seid test involves switching off the DUT rapidly, disrupting the current flow and causing a rapid increase in voltage (dv/dt) across the device because the stored energy in the inductance is being rapidly released. This test is crucial in assessing how well the device can handle abrupt changes and high-energy transitions without entering the destructive latch-up state. Fig. 14 shows V_{GS} , V_{DS} and I_{DS} waveform examples of an UIS test, as well as the avalanche energy ($E_{avalanche}$) calculation. Table 5 summarizes the main failure modes for Si-IGBT/MOSFETs and SiC-MOSFETs, their corresponding stress tests, affected parameters, and key reliability insights.

IV. PACKAGE-LEVEL RELIABILITY

Even though the semiconductor chip is the main component in power devices, its performance and integration depend largely on the packaging. The packaging of a system ensures proper

electrical functionality and also provides external usability for the power device. It also provides mechanical stability and integrity, helps dissipate the heat generated by power usage, and protects the system from environmental factors such as pollution and humidity [84].

There are three main reliability aspects that a power module must fulfil: thermal, electrical and thermo-mechanical. Thermal reliability focuses on the packaging's ability to dissipate heat that is generated during a semiconductor's operation. Effective thermal management is critical, as overheating can significantly degrade performance and reduce lifespan [86]. The packaging must be designed to efficiently conduct heat away from the semiconductor chip, ensuring that any temperature rise is minimised. Electrical Reliability addresses the challenges posed by stray inductances and parasitic capacitances in the module. Stray inductances can impede the performance of wide-bandgap (WBG) devices, especially during high-speed switching, leading to voltage overshoots and increased Electromagnetic Interference (EMI) [87], [88], [89]. Parasitic capacitances can create unwanted pathways for high-frequency currents, further complicating module behaviour and potentially compromising device reliability [89], [90].

The reliability of the entire power semiconductor module system depends not only on its electrical and thermal management capabilities but also on its thermo-mechanical reliability. Therefore, the module's ability to withstand and function properly depends also on its ability to handle the mechanical stresses caused by thermal fluctuations. These fluctuations occur due to differences in the coefficient of thermal expansion (CTE) among the different materials used in the module. The CTE measures how much a material expands or contracts when its temperature changes. In a module, different materials with different CTEs are bonded together. During operation, temperature changes cause these materials to expand and contract at different rates, creating stress at the material interfaces. This stress can lead to mechanical failures such as cracks

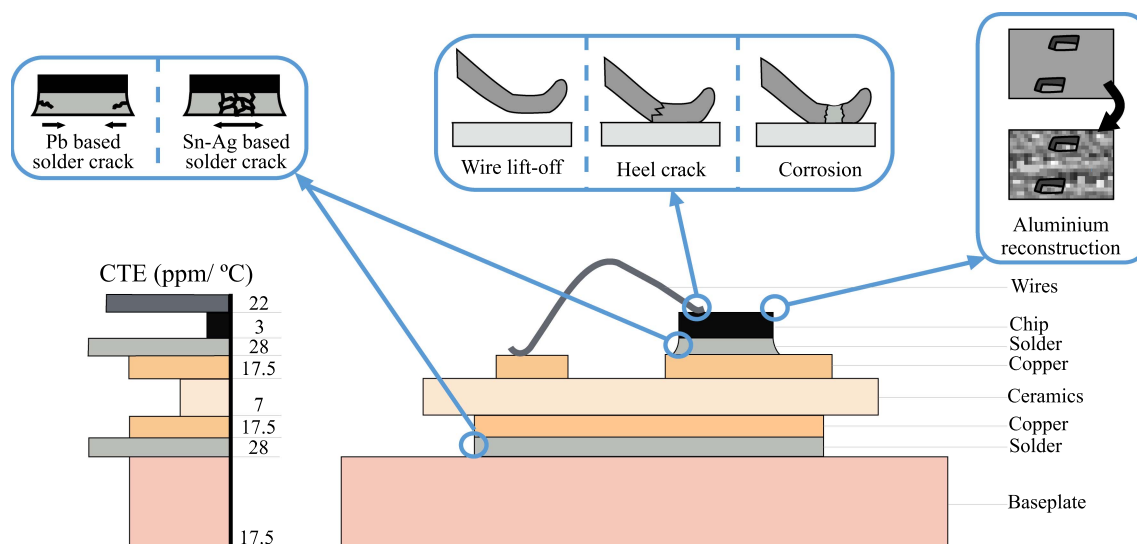


FIGURE 15. Structure and material's CTE of a standard power module and main failure modes. Adapted from [47], [54], [85].

and delamination, which can significantly affect the module's overall reliability [54].

Numerous studies have identified thermo-mechanical stress as the primary cause of power semiconductors' device aging, which corresponds to the Wear-out Failure zone previously mentioned (Fig. 2) [54], [85]. The different expansion rates of the materials used in these devices cause stress at multiple levels, resulting in a range of failure mechanisms. These mechanisms, such as bond wire lift-off, heel cracks, solder breakdown, and aluminium reconstruction, can negatively impact the integrity and operational lifespan of the semiconductor modules. Fig. 15 depicts the primary causes of failure that power modules commonly experience, while also indicating the CTE values for each material.

A. PACKAGE KEY ELEMENTS

To understand the performance capabilities of power modules, certain aspects must be identified, especially how a module is built; what its elements are; and how the purposes, features, requirements and actual solutions are defined. The traditional solution for IGBT and SiC-MOSFET power modules is a vertical structure. This is the preferred approach due to its numerous advantages, including the efficient conduction of high currents and the provision of high-power densities. The vertical structure is also resilient to thermal cycling variations and can efficiently evacuate heat flows. This is mainly due to the use of inorganic substrates like Direct Bonded Copper (DBC), which offers high-thermal conductivity, thereby reducing thermal resistance (R_{th}) and facilitating vertical heat dissipation from the semiconductors [20], [37], [84].

Additionally, the vertical architecture of power modules allows for easy integration with cooling systems, simplifying the assembly process by providing electrical insulation without the need for additional insulating layers. A conventional vertical structure module is typically composed of the following elements: power semiconductor chips, insulating

substrate, base plate, bonding material, power interconnections, encapsulant, and plastic case [47], [85]. An appropriate selection of materials is crucial for the construction of a durable package, including insulators, conductors, semiconductors, and both organic and inorganic substances; seeing that they all behave differently under various environmental, electrical, and thermal stresses [92]. Moreover, the price and ease of manipulating these materials during the package building process are also important factors to consider when manufacturing power modules. Having materials that are easy to work with can contribute to a more efficient manufacturing process, which in turn can ultimately lead to cost savings. It is therefore clear that the design of packaging consists in finding a compromise, and that the design is highly dependent on the application [90]. Fig. 15 shows how the key elements are arranged within the module.

1) BOND WIRES

Wire bonding remains the most widely used interconnection method in power modules due to its lower cost and simpler integration, especially when compared to press-pack technologies that, although more robust, require complex mechanical clamping [47], [93]. In power modules, two types of bond wires are typically used: gate wires, which carry control signals, and power-carrying wires, which handle high currents and contribute to heat dissipation [16]. The current-handling capability of bond wires depends on several factors, including the number of wires, their geometry, the type of current (AC or DC), and the thermal and electrical properties of the bonding material [94].

- *Aluminium wire bonding:* The most common method, formed by ultrasonic wedge bonding, favored for its manufacturability and reliability in standard applications [90].
- *Copper wire bonding:* Offers better thermal conductivity and lower electrical resistance, with a coefficient of

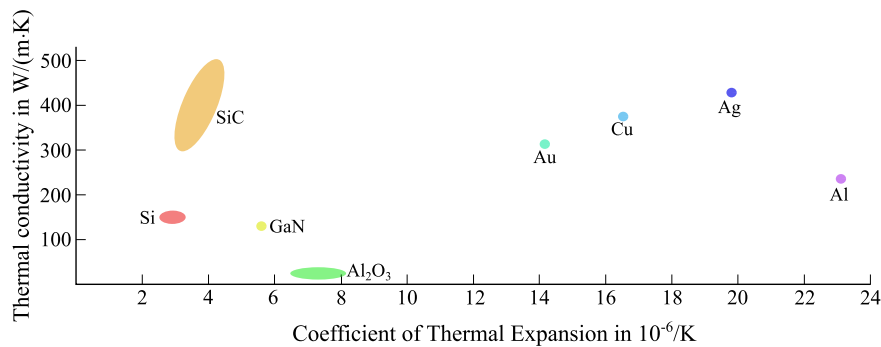


FIGURE 16. CTE values and thermal conductivity of different materials. Adapted from [91].

thermal expansion (CTE) closer to that of Si and SiC, reducing thermo-mechanical stress [95]. Fig. 16 illustrates the relationship between CTE and thermal conductivity for various bonding materials.

- *Ribbon bonding*: Provides lower parasitic inductance and improved power cycling performance, making it suitable for high-current applications [96].
- *Al/Cu wires (Al-clad Cu)*: Combine the thermal and electrical performance of copper with the bonding ease of aluminium [95]. However, they may require changes in die metallization and packaging processes, increasing complexity [97].
- *Advanced packaging strategies*: Technologies such as copper clips, direct lead bonding (DLB), copper post interconnects, or sintered copper blocks are being explored to eliminate bond wires altogether [37], [98]. These are already implemented in modules like RoadPaK and Tesla's Model 3 inverter to enhance short-circuit robustness and compactness.

Despite these advancements, aluminium wire bonding is still widely used in cost-sensitive applications due to its affordability and ease of processing. Ultimately, the selection of bonding technology is the result of a trade-off between cost and performance, and improving bond wire reliability alone is insufficient unless accompanied by enhancements in die-attach materials and design, which play a critical role in managing junction temperature and overall module robustness [95], [99].

2) JOINING (DIE-ATTACH)

Die-attach materials play a key role in ensuring the mechanical integrity, thermal management, and electrical performance of power modules. These materials connect the semiconductor dies to the substrate and the substrate to the baseplate, creating the necessary interfaces for efficient heat transfer and current conduction under elevated thermal and electrical stresses.

- *Sn-based solders*: Traditionally, Sn-based soft solders, such as SnPb or SnAgCu, have been widely used in this role due to their process simplicity and acceptable performance under moderate conditions [37]. However, these materials exhibit significant degradation near their melting point (221 °C), which leads to solder fatigue and

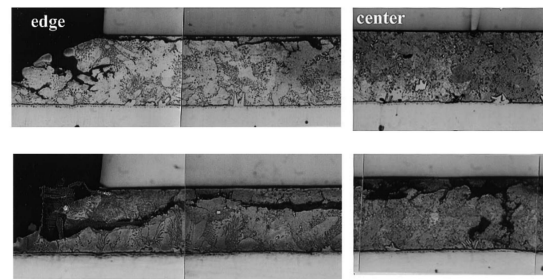
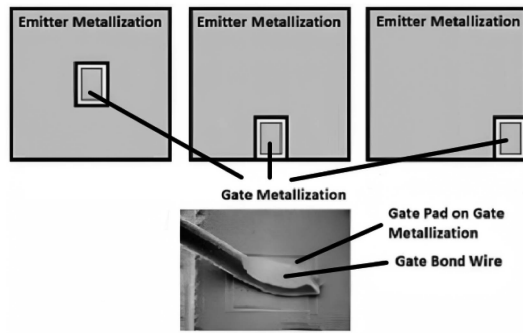


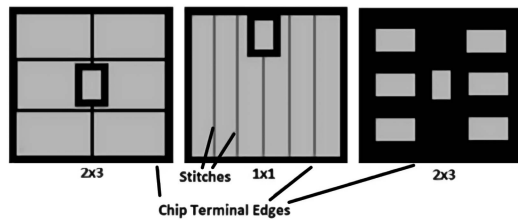
FIGURE 17. Optical micrographs of the cross section of the solder crack propagation after power cycling [100].

crack propagation during thermal cycling, as shown in Fig. 17 [100]. This behaviour restricts their reliability in applications requiring junction temperatures above 175 °C, especially with the emergence of wide-bandgap devices such as SiC [101].

- *Silver sintering*: In recent years, silver sintering, or Low-Temperature Joining Technology (LTJT), has emerged as an alternative that offers significant advantages over traditional soldering methods, particularly for high-temperature applications where high thermal conductivity is required [102]. It provides a more robust and rigid joint, improved thermal conductivity, and a reduced risk of solder layer failure, making it increasingly popular in demanding applications [103]. Silver sintering has a melting temperature around 960 °C, which is significantly higher than that of standard soldering, supporting its suitability for high-temperature operation [94]. Silver sintering also avoids the formation of intermetallic compounds, as the bonding mechanism relies on atomic diffusion at relatively low processing temperatures (230–260 °C) [104]. Commercial formulations, such as Heraeus' LTS 275-3P2 paste, have been successfully used for both chip-to-substrate and substrate-to-baseplate interconnects [105], [106].
- *Other techniques (TLPB and diffusion soldering)*: Transient Liquid Phase Bonding (TLPB) and diffusion soldering also offer promising reliability for high-temperature applications [107]. TLPB, in particular, creates a stable, high-melting-point bond (over 500 °C)



(a)



(b)

FIGURE 18. Metallization layouts [16]: (a) Gate pad layouts (centric, edge, and corner), including a detailed view of the gate metallization with a bond wire in place; (b) Common emitter metallization patterns.

through a metallurgical reaction between low- and high-melting-point components [105].

However, it is important to note that these advanced methods typically require adjustments to the metallization layers of the die and substrate to ensure compatibility, which can increase manufacturing complexity and cost [97]. Despite this, their reliability advantages make them increasingly attractive for SiC and next-generation high-power modules.

3) SEMICONDUCTOR SURFACE METALLIZATION

Semiconductor surface metallization, typically applied to the top and bottom surfaces of semiconductor devices, is used to establish robust electrical connections and effective thermal management. The top-side metallization, which includes gate and emitter pad-layouts, is essential for wire-bonding connections that link the semiconductor's active regions to the external circuit. On the other hand, the back-side metallization forms a thermal and electrical interface with the substrate. The metallization layers serve as a pathway for electrical conduction, and they also allow for the handling of overcurrent and short-circuit energy by means of their significant heat capacity [94].

Fig. 18(a) shows various layouts for gate pads (centric, edge, and corner) and a detailed view of gate metallization with a bond wire in place. The layout of the gate pad is determined by the bond-wire scheme used in the design of the power module, as it has a significant impact on the semiconductor chip's switching performance [16]. Emitter metallization designs typically feature multiple bonding pads

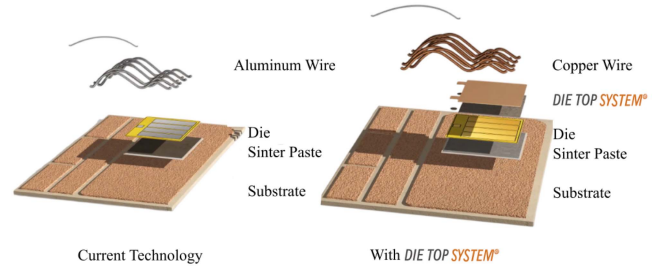


FIGURE 19. Comparison between traditional solutions and DBB technology in terms of packaging elements [106].

arranged in a matrix formation to prevent current crowding and hot spots, as shown in Fig. 18(b). These layouts are designed to ensure stable current distribution across the emitter metallization.

Metallization degradation can significantly impact device performance, leading to increased electrical resistance and compromised heat transfer. Additionally, this degradation may cause non-uniformity of the current density, resulting in bond-wire contacts failing. As a consequence, the remaining bond wires carry the full load with a higher current density. This current density may continue increasing until reaching a critical value [54].

With copper's superior thermal and electrical properties, copper metallization is an attractive alternative that can increase short-circuit capability and improve overall module performance. Innovations such as Danfoss Bond Buffer (DBB) technology replace the traditional solder joint between the substrate and semiconductor with a sintered joint and introduces a copper plate sintered on top of the semiconductor surface metallization (see Fig. 19). Utilizing a copper plate facilitates the use of copper wires, which in turn enhances current-density distribution, reduces ohmic losses, and improves thermal performance; thereby contributing to a higher power cycling capability. Moreover, authors in [108] increased critical short-circuit energy by 20–25%, replacing thin aluminium metallization with thick copper metallization, which acts as a temporary heat sink.

4) SUBSTRATE

In power modules, substrates are used as a base for mounting and interconnecting semiconductor devices. The primary function of the substrate is to establish a dielectric barrier, this way preventing electrical short-circuits between live components and the heat sink. It also serves as a heat spreader, moving thermal energy away from active devices to prevent overheating. Common materials for substrates include ceramics such as alumina (Al_2O_3) or aluminium nitride (AlN), chosen for their high-thermal conductivity and matching CTE to the semiconductor material. Direct Bonded Copper (DBC), Active Metal Brazing (AMB), and Insulated Metal Substrate (IMS) are among the typical substrate technologies employed in power modules. However, because of the remarkable ratio

of performance-cost, DBC aluminium oxide ceramic is the most common substrate [104].

In DBC, the ceramic is sandwiched between two conductive copper layers. The top-side copper layer provides a surface for electrical connections and thermal conductivity to spread heat to both the ceramic substrate and the bottom conductive layer. This last layer helps the ceramic substrate conduct heat vertically down to the heat sink, as copper has better thermal conductivity than the ceramic layer (Al_2O_3 : $24 \text{ W}/(\text{m} \cdot \text{K})$, Cu: $394 \text{ W}/(\text{m} \cdot \text{K})$) [12].

Under high-temperature cycles, DBC ceramic cracking and DBC metallization delamination are commonly observed. One of the primary reasons for substrate failures is the mismatch of coefficients of thermal expansion (CTEs) among different materials. Due to the CTE mismatch between copper metallization and ceramic material (Al_2O_3 : $7 \text{ ppm}/^\circ\text{C}$, Cu: $16 \text{ ppm}/^\circ\text{C}$), DBC technology fails under extreme temperatures, leading to mechanical stress during temperature variations [94]. New substrates are also evolving toward thinner structures, for both conductive layers (lead-frame technology) and ceramics (thin-film insulation technology), thus improving the thermal behaviour of the structure and facilitating component integration [20].

5) TERMINALS

As semiconductor technology advances, the ratings of current and power densities of modules increase while their size shrinks. However, the terminals of older standard packages are reaching their limits in carrying capacity, which requires limiting the current to prevent terminal heating. The specification of the maximum current flowing through the terminals, is measured in $I_{t(RMS)}$. In addition to the chip losses, losses in the terminals must be taken into account using $r_{cc'-ee'}$, which usually is not negligible. $r_{cc'-ee'}$ represents the sum of resistances of the bond connections and terminals. This parameter is sometimes provided in the datasheets. For instance, a module with $r_{cc'-ee'} = 0.5 \text{ m}\Omega$ would generate terminal losses of approximately 80 W at 400 A [12].

The carrying capability of currents is significantly lower for high-bonded terminals that are filled with compound in packages (Econopack). This is because there is only a poor thermal connection to the DBC substrate in these cases. On the other hand, the technology has an advantage in the mechanical connection of the terminals. This makes them more robust and resistant to external forces and moments, such as torque and bending moments [12].

If the $T_{j(max)}$ of the chip remains below its maximum limit during an overload, the terminals may become a bottleneck because the maximum current capability of the terminals ($I_{t(RMS)}$) is independent of the chips' junction temperature. Therefore, it is crucial to evaluate both $T_{j(max)}$ and $I_{t(RMS)}$ to ensure optimal performance during an overload. Fig. 20 shows an example where the nominal current of the module is 35 A for continuous operation. The $I_{t(RMS)}$ is 30 A per pin, and since there are two pins, the maximum terminal current

Absolute Maximum Ratings			
Symbol	Conditions	Values	Unit
Inverter - IGBT			
V_{CES}	$T_j = 25^\circ\text{C}$	1200	V
I_C	$\lambda_{\text{module}} = 0.8 \text{ W}/(\text{mK})$ $T_j = 175^\circ\text{C}$	41	A
	$T_s = 70^\circ\text{C}$	34	A
I_C	$\lambda_{\text{module}} = 2.5 \text{ W}/(\text{mK})$ $T_j = 175^\circ\text{C}$	50	A
	$T_s = 70^\circ\text{C}$	41	A
I_{Cnom}		35	A
I_{CRM}		70	A
Module			
$I_{t(RMS)}$	$\Delta T_{\text{terminal}} \text{ at PCB joint} = 30 \text{ K, per pin}$	30	A
T_{stop}	module without TIM	-40 ... 125	$^\circ\text{C}$
V_{test}	AC, sinusoidal, $t = 1 \text{ min}$	2500	V

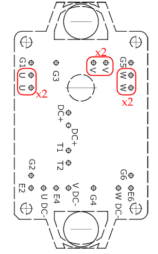


FIGURE 20. Datasheet information regarding terminal current [33].

is 60 A (averaged over a full operating cycle) [12]. However, considering the limited availability of the parameter $I_{t(RMS)}$ in datasheets, evaluating the terminal current capability can be challenging. Moreover, in most cases the $T_{j(max)}$ will remain the most critical aspect.

6) ENCAPSULANT

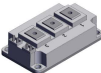
Encapsulants protect the integrity of the device, as they function as insulators to prevent electrical shorts, safeguard against environmental contaminants such as moisture and dust, and assist in managing thermal stresses in power modules. The most commonly used material for encapsulation is silicone gel, due to its excellent electrical properties, ease of processing, high elasticity, and self-healing properties [109]. Encapsulants are materials that can break down over time due to various factors such as thermal cycling, moisture ingress, or exposure to high voltages. These factors can cause partial discharges (PD), which are localized dielectric breakdowns that in turn can lead to material degradation over time. Encapsulants are particularly susceptible to PD, which accelerates aging and ultimately may lead to failure.

When selecting encapsulants, material compatibility within the power module package is crucial. Encapsulants must be compatible with other materials such as substrates, solders, and conductive adhesives in order to ensure overall package integrity. Incompatibilities can lead to delamination, void formation, and other failures.

For high-voltage applications that exceed operating temperatures of 200°C , choosing encapsulants which can withstand such conditions while maintaining excellent insulation and thermal properties is challenging. Advanced encapsulants may contain fillers or additives to enhance their performance, such as improving thermal conductivity or increasing mechanical strength. These improvements help to manage the increased power densities and operating temperatures seen in high-voltage, high-power applications [90], [94].

However, understanding the behaviour and limitations of encapsulant materials under high-temperature conditions remains a challenge. Efforts are being made to model their behaviour, but further research is required to entirely comprehend the complexities involved [110], [111].

TABLE 6. Overview of Standard Commercial Packages

	EASY	62 MM	ECONO	IHM/IHV	PrimePACK	XHP
						
	Flexible and scalable, base-plate-less design with improved thermal management	Reliable performance for superior thermal handling	Optimized for thermal stability and reliability in medium-power applications	Very high reliability in high-power, high-voltage environments	Scalable design, long-term performance oriented thermal management	Flexible, extreme environments up to 6.5 kV oriented high-reliability design
Infineon	SiC and IGBT	SiC and IGBT	IGBT	IGBT	IGBT	IGBT and SiC
Semikron	SiC and IGBT (Semitop)	SiC and IGBT (Semitrans)	IGBT (Semix)	-	-	-
Vincotech/Mitsubishi	SiC and IGBT (Flow)	SiC and IGBT (T series)	SiC (NX) and IGBT (VINco)	IGBT (STD type)	-	IGBT (LV100 type)
Wolfspeed	-	SiC (Wolfpack)	-	SiC (BM2)	-	SiC (LM)
Fuji	IGBT (Small PIM)	SiC and IGBT (62 mm)	IGBT (EconoPIM, EconoPACK)	-	IGBT (PrimePACK)	IGBT (HPnC)
ROHM	-	-	SiC (G type)	-	-	-
STMicro	IGBT and SiC (AcePACK)	-	-	-	-	-
ABB	-	IGBT (62Pak)	IGBT (LoPak)	IGBT (HiPak)	-	IGBT and SiC (LinPak)
Hitachi	-	-	-	IGBT (IHM)	-	SiC and IGBT (LV-nHPD2)
Toshiba	-	-	SiC (2-153A1A)	-	-	IGBT (XP1V)
Onsemi	SiC F family (F1 and F2)	-	SiC (PIM) and IGBT (SSDC)	-	-	-

B. STANDARD POWER MODULE PACKAGES

Table 6 outlines the main packages available in the market, highlighting their reliability-related characteristics. Although many package designs have become standardized across the industry, different manufacturers may use different names for the same configurations. The table includes the principal types of packages along with their names, as well as the chip technology used (Si-IGBT or SiC-MOSFET). Infineon's package families, used as standards in this analysis, showcase technologies that enhance performance, thermal management, and durability under varying operational conditions.

- **EASY Family:** Known for its simplicity and adaptability, the EASY family incorporates a base-plate-less design that minimises mechanical stresses and enhances power output. Its expanded DBC surface area provides better thermal conductivity, directly improving the reliability of modules under high thermal loads. These features, combined with support for custom layouts, make the EASY family highly resilient for high-volume production and long-term operation in industrial drives and renewable energy systems.
- **62 mm Modules:** Designed for medium-power systems, these modules support both IGBT and CoolSiC™ technologies, offering robust power cycling capabilities. Their integration of advanced IGBT7 and CoolSiC™ MOSFETs ensures reduced $R_{DS(on)}$, which directly enhances thermal performance and extends

device lifespan. With configurations optimised for low inductance and effective thermal management, these modules address reliability concerns for inverters and UPS systems operating in dynamic environments.

- **Econo Family:** Compact and cost-effective, the Econo family focuses on reducing parasitic inductance and enhancing thermal stability. Copper base plates ensure efficient heat dissipation, directly contributing to the longevity of power modules. Their thermal and mechanical reliability makes them ideal for medium-power applications such as industrial drives.
- **IHM/IHV:** Designed for high-power and high-voltage applications, these modules prioritize thermal stability and robust construction. Their ability to handle up to 3600 A and voltages of 4.5 kV ensures reliable performance in demanding environments such as rail traction and heavy industrial equipment. Features like optimised switching losses and superior power density enhance their operational resilience.
- **PrimePACK:** Developed for modular and scalable designs, PrimePACK modules prioritize long-term performance and thermal management. Their flexible footprint and compatibility with multiple IGBT generations allow for reliable upgrades and system scalability in energy systems like wind and solar converters.
- **XHP Family:** Built for ultra-high-power applications, the XHP family focuses on design flexibility and

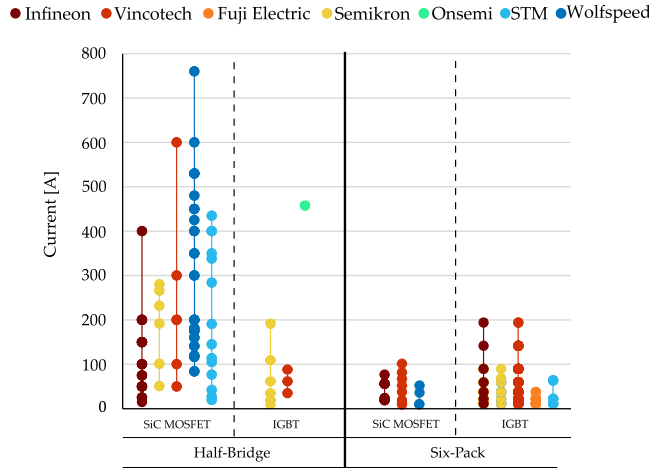


FIGURE 21. Commercial Easy package SiC-MOSFET and IGBT at 1200 Voltage Class.

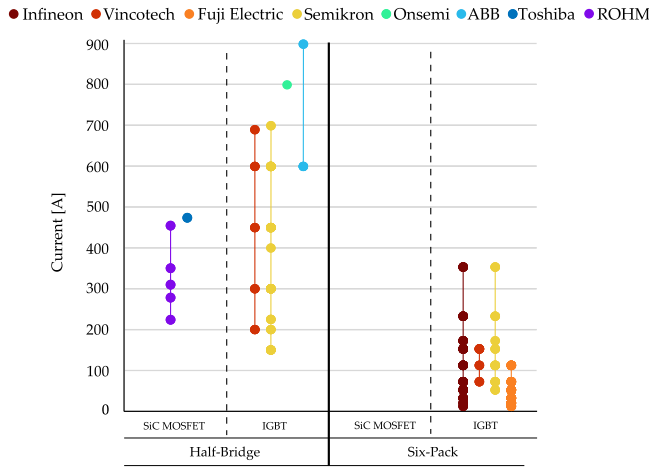


FIGURE 22. Commercial Econo package SiC-MOSFET and IGBT at 1200 Voltage Class.

reliability in extreme environments. Standardized housing dimensions and compatibility across voltage ranges up to 6.5 kV enhance resilience and adaptability in traction, commercial vehicles, and medium-voltage drives.

In terms of the operating ranges of the available power modules being marketed by various manufacturers, Fig. 21 displays a classification of the power modules with EASY family packaging (Half-bridge and Six-pack) offered by leading semiconductor manufacturers, categorized by current ranges and for 1200 Voltage Class. Conversely, Fig. 22 presents the same analysis for Econo packages. It is worth noting that there are currently no commercial six-pack solutions for SiC technologies with Econo package.

These two examples are intended to provide representative insights into current trends in packaging for Si and SiC technologies, particularly in terms of current capability and commercial availability. Their purpose is not to offer a comparative analysis of all package families in Table 6, which

would require a broader dataset and a dedicated case-by-case assessment beyond the scope of this review.

V. ACTIVE THERMAL CONTROL

Active Thermal Control (ATC) strategies are developed to improve the thermal performance of power electronic devices by controlling junction temperature cycling. ATC can significantly mitigate thermal stress by modulating critical operational parameters such as gate voltage, output power, or switching frequency among others [17], [85], [101], [112], [113], [115], [116], [117]. These manipulations help maintaining the maximum junction temperature ($T_{j(max)}$) within safe limits, reducing the average operating temperature ($T_{j(av)}$), and minimising temperature fluctuations.

It is important to highlight that while chip-level and package-level designs are typically defined at the manufacturer level, they remain less adaptable for modifications by end-users. In contrast, ATC techniques can be directly implemented by end-users of power modules, such as electric vehicle manufacturers, power grid operators, and fast-charging infrastructure developers. These techniques can significantly impact the overload capability and longevity of the devices. However, it is important to analyse not only the efficiency, performance, and lifetime improvement of the control strategies, but also the overall system cost, the drawbacks and the feasibility of their integration into existing applications. This ensures that solutions are not only technically effective but also economically viable and easy to adopt at real-world application levels. ATC can be beneficial not only during standard conditions but also during overloads. As a result, power converters may safely be pushed to their maximum thermal limits.

ATC techniques can be classified in various ways. A common approach is based on the hierarchy of implementation: device level, converter level, and system level (see Fig. 23(a)) [27], [112]. At the device level, the stress on each device in the converter is individually regulated. Then, to the converter level, and finally to the system level where a converter in a system of multiple converters is treated as a single block [112]. Some authors also add a fourth level corresponding to the modulation level [113]. Even a fifth level could be considered, namely the control of heat dissipation or cooling [101].

Another way to classify ATC techniques depends on the control objective: increasing overload capability, thermal cycle reduction, balancing thermal stress of power devices, and thermal stress redistribution [112], [114], as shown in Fig. 23(b). Increasing overload capacity focuses on enhancing the ability of power devices to handle higher than normal loads without compromising reliability. By dynamically adjusting parameters like switching frequency or gate voltage to reduce the losses, devices can operate closer to their thermal limits.

Thermal cycling reduction is essential to the longevity of power devices due to the thermo-mechanical stress. Reducing the amplitude of temperature fluctuations prolongs device life

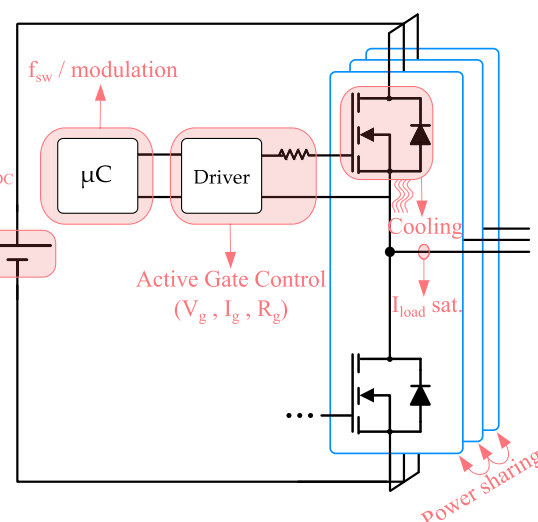


FIGURE 23. Identification of: (a) hierarchy of implementation [27], [101], [112], [113]; (b) ATC technique objective [112], [114]; (c) main ATC techniques and where are they applied.

by mitigating thermo-mechanical fatigue, and can be achieved by modifying the device's operating conditions to smooth out the temperature profile across operational cycles. In systems with multiple power devices, such as multilevel converters or multichip modules, thermal stress can be unevenly distributed, thus leading to overheating and premature failure of individual components. ATC can be employed to balance the thermal load across all devices, ensuring uniform temperature distribution and enhancing the overall reliability and efficiency of the system.

Latter objective involves strategic load distribution among multiple converters within a system to optimize the thermal stress based on the aging and health of each component. It aims to prolong system life by actively managing which parts of the system bear more stress at any given time, based on real-time assessments of component health and performance. This strategy can lead to more uniform aging across the system, facilitating more predictable maintenance schedules and potentially reducing overall maintenance costs. Although there are some similarities with thermal stress balancing, balancing is about maintaining equality to prevent stress concentration, while redistribution focuses on strategic load placement to optimize system longevity based on component conditions. Fig. 23(c) identifies the specific areas where each ATC technique is applied.

A. OVERVIEW OF ATC METHODS

1) ACTIVE GATE CONTROL

Active gate control enables the regulation of semiconductor losses without affecting the device's functionality. Different variables like gate voltage, gate current or gate resistance can be manipulated in order to achieve power loss reduction.

The gate drive voltage affects both, conduction and switching losses of the semiconductors, therefore enhancing their overload capability. By adjusting the control parameters in these active gate controls, the thermal cycling in the semiconductors is minimised [118], [119].

The gate voltage significantly impacts on the output characteristics and power dissipation of the device. In practice, many applications maintain a significant margin from the maximum value for various reasons such as gate-oxide reliability or maximum current capability of the module among others. However, even a small increase in voltage, such as going from 15 V to 18 V, can lead to significant differences in on-state resistance and losses for certain device types. For example, the datasheet output characteristics of a 3.3 kV IGBT module show that a change from 15 V to 20 V gate-driver voltage amplitude can result in a 1 kW reduction in on-state power losses for a reference current value of 2 kA [120].

A two-level slew-rate control driver is presented in [121], aiming to optimize IGBT performance by adjusting the gate resistor value according to operating conditions and efficiently accommodating different load scenarios. The study demonstrated that this approach reduces switching losses, limits dv/dt levels effectively, and thereby manages to maintain lower junction temperatures across various operational states.

In [122], increasing the gate drive of the DC-DC converter from 15 V to 20 V improved the inverter's efficiency, rising from 97% to 97.5%. In addition, [81] demonstrates the impact of different V_G and R_G values in the device lifetime according to real field mission profiles. For instance, at a high V_G of 20 V (resulting in low conduction losses), the increment in R_G has a negative impact, reducing the device lifetime by up to 50%, from 20 years ($V_G = 20$ V and $R_G = 7$ Ω) to 9.85

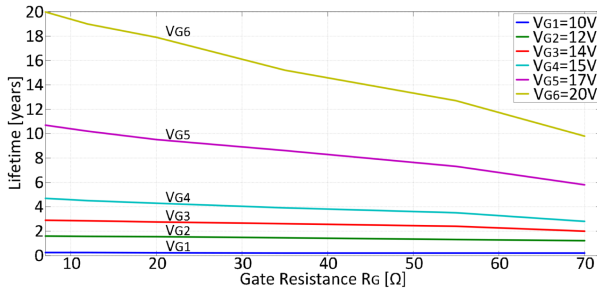


FIGURE 24. Impact of variations in V_G and R_G on the lifetime of power semiconductors in PV inverters [81].

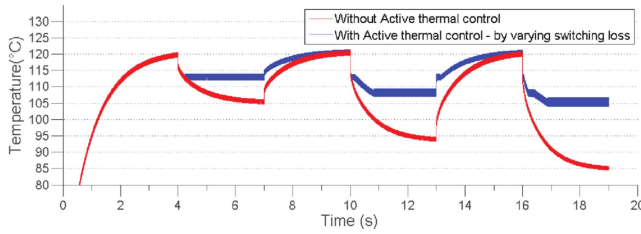
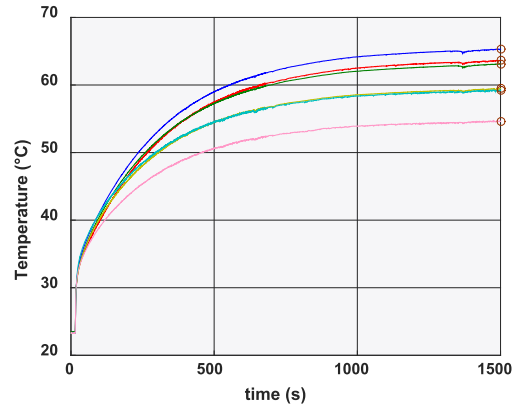


FIGURE 25. Examples of ATC based on Active Gate Control to reduce the temperature ripple by increasing power losses [126].

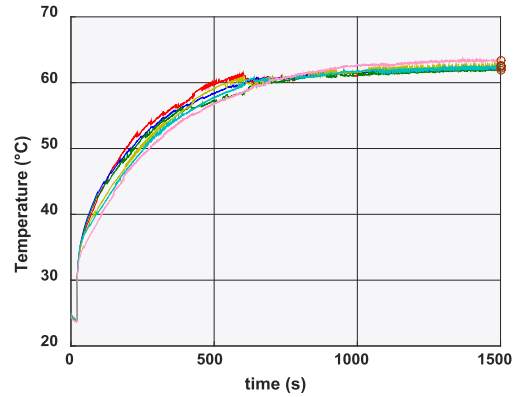
years ($V_G = 20$ V and $R_G = 70$ Ω). The lifetime results shown in Fig. 24 were obtained for different V_G and R_G values. It is evident that the reduction of conduction losses (considering the output characteristics of the device) or switching losses (decreasing the switching time by modifying the slew-rate when modifying gate resistance or gate voltage/current), significantly increases the lifetime of the device. Additionally, the overload capability is enhanced, as the device can withstand harder working conditions at the same junction temperature.

Another benefit of the active gate control technique is that, unlike other techniques, it does not affect the current ripple of the converter output. However, gate control requires hardware modifications and additional circuitry, therefore posing a significant drawback for its implementation and final adoption. Moreover, the end user should be made aware of the fact that when increasing the gate voltage, a reduction in short circuit withstand time (t_{psc}) occurs [123], [124], [125]. One alternative way to address these issues is to increase the gate voltage only during the conduction phase, thereby switching at the nominal gate voltage value. However, this solution requires a fast response time capable of varying the gate voltage level based on the device's state. Additional details about this aspect are presented in Section V-B, where the impact of increased gate voltage on the short-circuit withstand-time is explained.

Some scientific articles, like [126], [128], use variations in gate voltage levels to reduce temperature ripple by increasing the $T_{j(min)}$ value, when the device is cooling down, as shown in Fig. 25. To do so, the power losses are usually increased during the cool down phase, which negatively impacts the



(a)



(b)

FIGURE 26. Example of ATC based on Active Gate Control to balance the temperature across six parallel devices: (a) without ATC technique, (b) with ATC technique [127].

total power consumed by the converter. In some cases where thermal stress is balanced between parallel semiconductor switches, the $T_{j(max)}$ is also increased [127]. Fig. 26 presents examples of ATC techniques based on active gate control for both, temperature ripple reduction by increasing power losses and temperature balance across six parallel devices. However, increasing the minimum junction temperature is not a viable strategy when aiming to improve overload capability. This approach reduces the available margin for potential overload situations.

2) SWITCHING FREQUENCY F_{SW} VARIATION

Another important ATC technique involves adjusting the switching frequency to manage thermal loads, especially during periods of high operational demand. By dynamically adjusting the switching frequency, it is possible to reduce thermal stresses within the semiconductor devices, which in turn affects their operational and thermal behaviours. For example, reducing the switching frequency during high load conditions can significantly decrease switching losses, thereby reducing peak junction temperatures and minimising thermal cycling experienced by the devices [117].

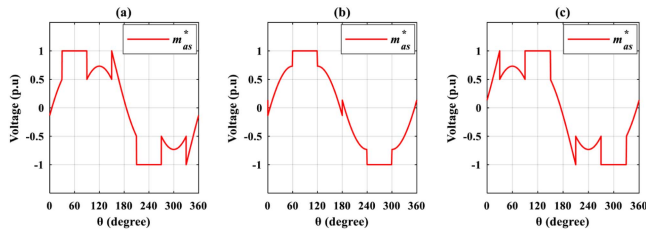


FIGURE 27. Different DPWM modulation signals with a 60° clamping (for one of the three phases): (a) DPWM0, (b) DPWM1, (c) DPWM2 [112].

In the case of frequency variation analysis, similar to active gate control, many authors increase the frequency to reduce temperature fluctuations by increasing the minimum junction temperature [129]. As mentioned previously, increasing the $T_{j(min)}$ is not suitable for improving overload performance, despite being useful for reducing lifetime consumption as the temperature fluctuations are smaller.

One of the main drawbacks of this approach is that the passive components need to be designed for the worst case scenario, resulting in an increased converter volume [17]. Moreover, the current total harmonic distortion (THD) increases almost linearly with reduced switching frequency, worsening the quality of the output signal [130]. For many applications, such as grid-connected systems, this limitation could be significant due to the strict THD requirements at the point of common coupling [112]. Therefore, these ATC strategies must exclusively address these limitations.

3) MODULATION TECHNIQUE

Another method for ATC involves manipulating the modulation techniques of the converter's gate signal. Different pulse width modulation (PWM) strategies have different impacts on the converter device loading, which means that they can be used as a manipulation variable for ATC.

In [131] a method is proposed to minimise thermal stress on semiconductor modules using discontinuous pulse-width modulation (DPWM). This method clamps the output voltage to the positive or negative DC link voltage during a specific interval to reduce the switching loss, resulting in total loss reduction during the clamping period. As a result, the average junction temperature and thermal cycling are reduced. Fig. 27 shows three different DPWM modulation signal possibilities with a 60° clamping (for one of the three phases). Changing between continuous and DPWM to reduce losses was also applied in [132], but careful consideration needs to be given to the application because of the different impacts on the current ripple.

While DPWM strategies can improve thermal management and efficiency, they also need to be carefully integrated into the system to maintain or improve the quality of power output. Adjustments in modulation strategies must consider the overall impact on system performance, including potential changes in the harmonic content and ripple in the output currents and voltages [112].

4) LOAD CURRENT I_{LOAD} SATURATION

The load current I_{load} limitation strategy in ATC focuses on regulating the output current to manage the junction temperature. This approach reduces the output current when a predetermined thermal threshold is exceeded [133].

The technique can be used as a saturation solution to avoid overheating, but it does not reduce the power losses of the device or help increase the power capability. The main drawback is that the output current is limited, making the device unable to withstand overload events. This limitation restricts the output current and could potentially limit the power capability of the device during peak demand scenarios. This makes it less suitable for applications where constant high power output is crucial.

5) DC-BUS VOLTAGE V_{DC} VARIATION

This technique is a strategy that manages thermal stress in systems equipped with DC-bus (V_{DC}). By dynamically adjusting the V_{DC} in response to changes in motor speed or operational demand, this technique effectively reduces power losses in the inverter, leading to decreased thermal stress on the components. Common control strategies vary the V_{DC} as a function of the motor speed [134].

In [135], dynamic adaptation of the bus voltage from 290 V to 750 V matches the torque-speed requirements of the operation. This led to substantial reductions in motor and inverter losses compared to a fixed DC link voltage of 550 V. This adjustment also resulted in a reduction of about 10 in temperature ripple during an acceleration test from 0 to 1000 RPM. Nevertheless, the bus voltage adaptation technique was combined with I_{load} saturation technique, which, in the end, caused a small reduction in the acceleration.

Using the V_{DC} variation technique requires careful calibration to prevent voltage adjustments that could damage sensitive components. Incorrect settings or control system failures can lead to unstable operating conditions, which may increase the risk of random failures.

6) POWER SHARING

If the system is composed of multiple cells or multiple converters, additional high-level controls can be implemented [116]. Power sharing is an ATC technique used to manage thermal load and extend the Remaining Useful Life (RUL) of converters within a system. This strategy redistributes power among converters to achieve balanced thermal distribution or equalize the RUL during operation, which is crucial in systems like microgrids, parallel DC-DC converters in aircraft, wind energy applications, and multi-phase drives. For example, in systems with parallel-connected converters, such as interleaved DC/DC converters, power routing is conducted by dynamically adjusting the power allocation based on the calculated damage and current ratings of each converter. This proactive management helps maintain operational integrity and longevity of the entire system by ensuring that no single converter is overstressed [27].

Additionally, reactive power sharing is another facet of this strategy, especially useful in grid-connected and multi-converter scenarios [136]. This method involves manipulating the reactive power generated and absorbed by different converters in the system. By controlling the reactive power, converters can be operated under varied thermal conditions, which is beneficial mainly during low-load situations. While the power sharing strategy offers significant benefits in balancing thermal loads and extending the Remaining Useful Life (RUL) of power converters, it has specific limitations that might affect its utility in some applications, particularly those requiring high overload capability. The main drawback of this technique is that it primarily redistributes power among converters without reducing the overall system losses. This redistribution aims to balance wear and thermal stress but does not inherently decrease the thermal load each converter must handle.

Unlike strategies that directly reduce the operational temperature or modify operational parameters to decrease loss, power sharing does not enhance the converters' ability to handle overload situations. It ensures that no individual converter operates beyond its thermal limit by distributing the load more evenly, but it does not increase the system's capacity to handle higher loads beyond designed specifications [112].

7) COOLING SYSTEM VARIATIONS

ATC strategies employing cooling system variations involve adjusting the cooling power dynamically to manage junction temperatures in power semiconductors. This strategy uses variable control over cooling components, such as fans or pumps, to respond to changes in the thermal load of the system [17], [101].

In [137], an open-loop controller is used to monitor power losses and ambient temperature with enhanced dynamic response, while an additional closed-loop controller is employed to control the baseplate temperature. In [138], a PID compensator is tuned to the thermal impedance of the semiconductor and utilised to control the blower speed, enabling the generation of output based on power losses without requiring temperature feedback. Additionally, in [139], ambient temperature is considered when controlling the cooling power to maintain a constant temperature.

The chosen strategy may lead to higher average temperatures compared to uncontrolled systems, which may not be desirable in all applications due to increased thermal stress during peak operations. For instance, in [139] the temperature ripple is almost eliminated, but the average temperature of the device increases by 45 and the maximum junction temperature also rises, as shown in Fig. 28(a).

The interpretation of this fact can vary depending on the control strategy. For instance, a comparison can be made between a scenario where the fan operates at a constant speed and another where the fan's speed is variable, ranging from its minimum to maximum. This is because in some cases, it is not feasible to maintain the fan at its maximum speed. For

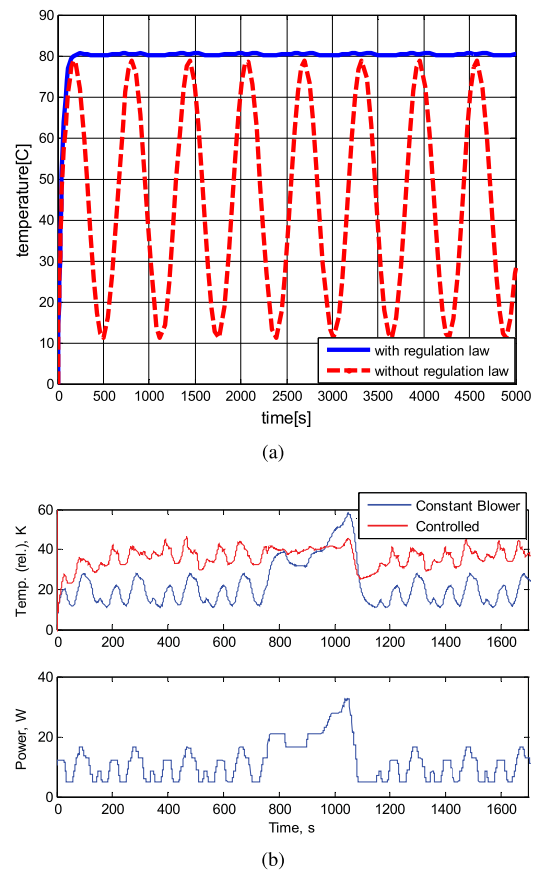


FIGURE 28. Implementation of cooling system variation (a) increasing temperature, and (b) regulating blower speed [138], [139].

example, in [138], a comparison is made between a constant speed of 2900 rpm and a variable speed up to 6000 rpm, as shown in Fig. 28(b).

8) INTEGRATION OF MULTIPLE ATC TECHNIQUES

While the ATC techniques discussed above are often studied in isolation, recent research has demonstrated the potential of combining multiple control strategies to enhance thermal performance more effectively. For instance, in [140], a combined control strategy based on switching frequency modulation and current limitation was implemented to regulate the junction temperature. In [141] and [114], a methodology that merges adaptive gate resistance with switching frequency adjustments is presented, selecting control values based on the estimated average junction temperature through thermal modelling and sensor feedback. Similarly, in [142], both gate voltage and gate resistance were jointly tuned to control switching losses using real-time junction temperature measurements.

Such hybrid techniques improve the accuracy and effectiveness of thermal control, but also introduce greater complexity in terms of control architecture and hardware implementation. Integrating multiple ATC strategies requires coordinated design to prevent conflicting control actions, and might demand

more sophisticated real-time temperature estimation and controller tuning [112], [114].

B. ADDRESSING RELIABILITY IN ATC TECHNIQUES

As ATC strategies become increasingly important for improving efficiency and extending the lifecycle of power electronic systems, it is crucial to ensure that these techniques are not only effective but also reliable and robust under various operating conditions. Ensuring the reliability and robustness of ATC implementations involves testing and validation against reliability tests.

This sub-section examines the standard tests mentioned throughout the document, providing details on how each test is relevant to verifying the performance and durability of ATC techniques.

1) SOA RELATED TESTS

In order to check the reliability and robustness of the ATC techniques, the following tests among the SOA can be performed. The Short-Circuit (SC) test ensures the short-circuit withstand-time (t_{pSC}) of the device for different conditions. For example, if the gate voltage wants to be increased, it is known that the short-circuit time will be reduced. Thus, the SC test provides information about these “new” limits for the new conditions. In terms of the SOA representation in Fig. 3, this test is related to the upper boundary, where high current are present. Some derating can be developed in order to estimate the impact of this variation.

For instance, one possible derating approach involves analysing the impact of increasing the gate voltage to reduce t_{pSC} . In this case, higher V_g values inherently lead to an increase in I_{SC} . The slope of the transfer characteristic can be used to calculate the factor for the increase in I_{SC} . According to [123], it can be assumed that the short-circuit energy (E_{SC}) of the device is constant, and can be calculated as shown in (6).

$$E_{SC} = I_{SC} \cdot V_{CC} \cdot t_{pSC} \rightarrow t_{pSC} = \frac{E_{SC}}{I_{SC} \cdot V_{CC}} \quad (6)$$

For example, if $I_{SC}(V_g = 15V) = 4 \cdot I_{C(nom)}$ and $I_{SC}(V_g = 17V) = 5 \cdot I_{C(nom)}$, and if $t_{pSC}(V_g = 15V) = 10\mu s$, then the new $t_{pSC}(V_g = 17V)$ can be calculated as follows:

$$t_{pSC}(17V) = t_{pSC}(15V) \cdot \frac{4 \cdot I_{C(nom)}}{5 \cdot I_{C(nom)}} = 8\mu s \quad (7)$$

It should be emphasized that this example serves as a theoretical estimation based on the constant short-circuit energy assumption ($E_{SC} = \text{const.}$), as introduced in [123]. While this method is useful for derating analysis, experimental validation is required to obtain accurate t_{pSC} values. For instance, in [143], short-circuit tests performed on 1200 V SiC-MOSFETs showed that increasing the gate voltage from 15 V to 18 V (at $V_{DC} = 600$ V) reduced the short-circuit time from 9.5 μs to 6.5 μs . This result is consistent with the trend predicted by the theoretical estimation and confirms the need

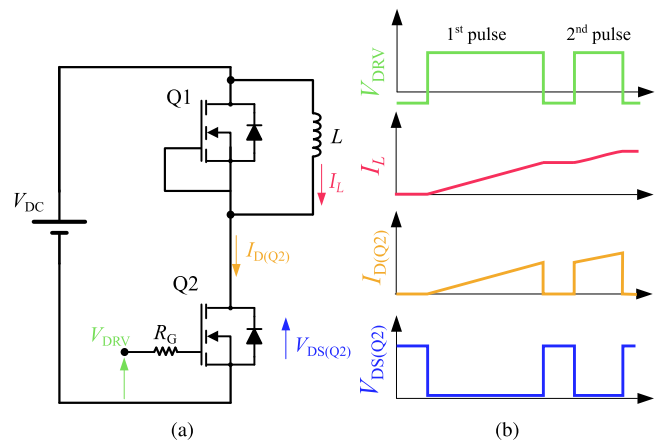


FIGURE 29. (a) DPT circuit configuration and (b) typical current and voltage waveforms.

to perform SC tests when adapting gate voltage conditions in Active Thermal Control strategies.

As mentioned in Section III-C4, the Unclamped Inductive Switching (UIS) test is useful for analysing the device’s avalanche capability. In the SOA diagram of Fig. 3, this test is associated with the high-voltage region, where avalanche stress may occur. The UIS is particularly useful to ensure that the critical limits along the voltage axis are not exceeded.

Additionally, in order to analyse the influence of the ATC technique on the switching behaviour of the device, the Double Pulse Test (DPT) can also be implemented. This test involves turning the DUT on and off twice in rapid succession to observe its behaviour. This test is illustrated in Fig. 29. When the device turns on, the current increases at a linear rate through a load inductor, which remains almost constant when the device is turned off. This produces a turn-off and turn-on event at the same current level, which is configured by controlling the length of the initial pulse. This test is widely used to determine differences in switching characteristics between turn-on and turn-off, such as rise and fall time or switching losses.

2) POWER CYCLING TEST FOR ATC

Power cycling tests can also serve as an indicator for performance enhancement. For the same mission profile, the different thermal responses can be analysed, thus concluding whether the ATC method has improved the performance or not. Moreover, extreme mission profiles with exigent turn-on power pulses can be tested until maximum temperature limits are reached. By doing so, the efficacy of ATC techniques is also tested not only to reduce lifetime degradation, but also to ensure that the device will survive such working conditions. The power cycling test is considered the most comprehensive to perform because it allows real-life mission profiles to be implemented into the DUT. Moreover, the test can identify different failure or degradation mechanisms that may not be detected in other static tests.

TABLE 7. Overview of ATC Techniques, Related Literature Works and Their Drawbacks

ATC Technique	Literature Works	Drawbacks/Challenges	HW/SW
Active gate control	[81], [121], [122], [126]–[128], [140], [141]	- Requires hardware modifications and additional circuitry - The device operates near the manufacturer's specified maximum voltage limits when the gate voltage is increased [120] - Gate resistance control cannot be effectively used with devices operated at high switching transient [112], [141]	HW
f_{sw} variation	[117], [140], [141]	- Usually used to increase $T_{j(min)}$ by increasing losses, which have an impact on the efficiency of the converter [17] - To limit the mean value of T_j, it is necessary to decrease the switching frequency, so passive elements must be designed for the worst-case scenario [17] - Current total harmonic distortion (THD) increases almost linearly with reduced switching frequency, worsening the quality of the output signal [112]	SW
Modulation	[131], [132]	- Many of the modulation methods used to reduce thermal stress have shown adverse effects on THD [112], [132]	SW
I_{load} limitation	[133], [140]	- Does not reduce the power losses of the device nor does it help increase the power capability - This limitation restricts the output current and could potentially limit the power capability of the device during peak demand scenarios	SW
V_{DC} variation	[134], [135]	- Risk of random failures may increase [112], [115]	SW
Power sharing	[116], [136]	- Applicable only on multiple cells or multiple converters - Reactive power circulation increases the power loss of all converters participating in the thermal stress reduction, which may lead to large power losses [112], [136]	SW
Cooling system	[137]–[139]	- Due to the inertia of the heat sink, it may not be possible to act within the range of microseconds or milliseconds - It may also provoke the oversizing of the cooling system.	SW

Traditionally, in standard power cycling tests, the failure criteria are the increase in thermal resistance, on-state resistance, and gate leakage current [17]. However, the evaluation of ATC techniques primarily focuses on their effectiveness in reducing thermal stress and enhancing overload capability across mission profiles. While the impact of degradation indicators is not the primary focus in such cases, it remains an important aspect for future investigations.

3) GATE-OXIDE RELIABILITY

Most of the reliability issues are common for all ATC techniques while others, such as gate-oxide reliability, must be checked only for active gate control based on increasing gate voltage.

However, as mentioned in Section III-C1, while gate-oxide degradation remains a concern, advancements in gate-oxide quality have significantly improved its reliability. Furthermore, in overload scenarios where active gate control should be applied only at specific moments, there is even more margin for such control.

It is important to highlight that the ATC technique based on gate voltage increasing will only be applied at specific moments when an overload must be withstood or when the

temperature ripple due to extreme load variation needs to be reduced. Therefore, the gate voltage will not be continuously maintained at this high voltage level.

C. ATC TECHNIQUE CONCLUSIONS

Table 7 summarized the main ATC techniques, their main limitations and the literature works that applied these techniques. Most ATC techniques are software-based, meaning they utilise existing hardware while employing control algorithms to mitigate thermal stress. This makes software-based thermal management methods highly cost-effective, as they typically require minimal additional hardware and can often be integrated into existing converter systems.

In contrast, ATC techniques based on Active Gate Control necessitate hardware modifications and additional circuitry, which poses a significant challenge for widespread adoption. However, their rapid actuation time and ability to effectively reduce power losses make them a suitable choice for high-overload and lifetime-critical applications.

Switching frequency variation and modulation techniques share a common drawback: they can impact output signal quality by increasing THD. Despite this, their rapid response times and effectiveness in reducing switching losses make

them viable options for applications requiring high reliability under overload conditions and where extended device lifetime is critical.

Regarding the I_{load} limitation technique, it primarily functions by limiting the maximum current value based on temperature. While it helps prevent overheating, it does not reduce power losses nor directly enhance overload capability without compromising device performance, which limits its applicability in scenarios requiring high thermal performance.

Power-sharing techniques, on the other hand, are specifically designed for systems with multiple converters or cells. These methods ensure that no single converter exceeds its thermal limits by distributing the load more evenly. However, they do not increase the overall system's capacity to handle higher loads beyond its designed specifications, which limits their utility for expanding performance boundaries.

The V_{DC} voltage regulation method shows promise due to its ability to improve overload capability and reduce temperature swings. Although it requires careful adjustments to avoid potential damage to sensitive components, it remains a suitable option for enhancing system reliability and thermal performance.

Lastly, cooling system-based solutions are constrained by the slow response times of heat sinks and their inherent thermal inertia. While this limits their applicability in scenarios requiring rapid thermal control, these solutions are not entirely dismissed. They can be strategically applied for temperature ripple reduction by dynamically adjusting fan speeds, making them useful for applications with specific thermal management needs.

VI. FUTURE CHALLENGES AND TRENDS

Recent discussions have emphasized the importance of chip technology, advancements in packaging, reliability testing, and active thermal control (ATC) in enhancing the performance and longevity of power semiconductor devices. However, several challenges still need to be addressed to promote the widespread adoption and further improvement of these technologies.

A. RELIABILITY TESTING CHALLENGES

Existing reliability methodologies have primarily been developed for Si-based technologies and do not fully address the unique degradation mechanisms of SiC semiconductors. As SiC becomes increasingly popular in high-performance applications, it is crucial to redefine reliability tests to align with its distinct characteristics.

One of the key challenges is to adapt standardized tests for SiC devices. Traditional power cycling and thermal cycling tests, widely used for Si-based devices, need modification to reflect SiC's different thermal properties and primary failure mechanisms. Additionally, commonly used chip-level tests, such as HTGB, HTRB, and H3TRB, must be reassessed to ensure they effectively capture SiC-specific degradation processes, including gate-oxide reliability issues and defect-driven failures.

Another critical challenge is to develop accurate lifetime curves. In many predictions regarding the lifetime of SiC-MOSFETs, data from lifetime curves originally developed for Si-based semiconductors are utilised. Therefore, it is essential to establish power cycling curves and lifetime estimation methods specifically tailored to SiC technology.

Additionally, while this article primarily addresses Si and SiC reliability, it is important to note that GaN technology is also evolving toward higher voltage applications. As GaN power devices are developed for medium and high-power applications, similar reliability challenges and testing adaptations will need to be considered in future studies.

B. CHIP AND PACKAGE TECHNOLOGY CHALLENGES

As demands for higher switching frequencies and elevated temperature operations increase, IGBTs are reaching their practical limits. As WBG devices, particularly SiC-MOSFETs, replace silicon-based solutions, several challenges arise at both the chip and package levels. SiC-MOSFETs carry unique reliability challenges compared to their silicon counterparts. Issues such as threshold voltage drift, gate-oxide degradation, and long-term stability under power cycling remain critical concerns that require further investigation.

Although IGBT technology has evolved significantly at the chip level, the main reliability bottleneck now lies in the package. For SiC devices, the challenge is even greater, as they still rely on packaging originally designed for IGBTs, which is not optimised for their higher switching speeds. Additionally, SiC's smaller chip size leads to higher power dissipation per unit area, worsening thermal management issues and reducing the short-circuit withstand time due to its lower inertia.

To overcome these limitations, packaging innovations are essential to fully leverage the benefits of both Si IGBTs and SiC-MOSFETs. Key advancements focus on:

- *Wire bond replacement*: Replacing bond wires with interconnection technologies like copper clips, ribbons, or silver clips will significantly enhance thermal capabilities [94].
- *Advanced die-attach materials*: Silver sintering and diffusion soldering provide stronger thermal and mechanical stability, reducing failure rates and increasing module lifespan [103], [144].
- *Improved metallization layers*: Thicker copper metallization enhances electrical conductivity and short-circuit robustness while also reducing ohmic losses. Techniques like Danfoss Bond Buffer (DBB) technology further improve power cycling capability [106].
- *Double-sided cooling power modules*: Double-sided cooling modules, which integrate heat-spreading structures on both sides of the semiconductor, significantly lower thermal resistance and improve temperature uniformity, enhancing the overall reliability of power converters [145].

C. ACTIVE THERMAL CONTROL CHALLENGES

While ATC has shown significant potential, several challenges still need to be addressed to ensure its widespread applicability and efficiency. Most existing ATC strategies have been developed and validated primarily for Si-IGBTs. However, SiC-MOSFETs require further investigation due to their unique failure mechanisms and thermal cycling behaviours. Additionally, ATC validation has often been limited to simulations, highlighting the need for more realistic experimental implementations that incorporate actual mission profiles and accelerated aging tests to evaluate long-term reliability improvements.

Another critical challenge is to minimise the impact of ATC strategies on overall system performance. Some ATC methods can introduce significant variations in power loss, which can affect converter efficiency and output signal quality. Moreover, interactions with other reliability-critical components, such as DC-link capacitors, must also be considered.

From an economic standpoint, ATC can significantly reduce maintenance costs by enabling predictive control of power semiconductor lifespan. However, the cost of implementation, including specialized hardware and control complexity, must also be carefully evaluated. Implementing smarter thermal management strategies can lead to reduced unscheduled maintenance, extended operational availability, and optimised safety margins, resulting in more reliable systems. Future research should focus on cost-effective ATC implementations that integrate state-of-health (SOH) monitoring and degradation-aware control strategies to enhance the longevity of power converters while maintaining economic viability.

Another fundamental aspect of ATC is the accuracy of junction temperature estimation. Many current sensing techniques suffer from low bandwidth, phase lag, or susceptibility to noise and degradation. Therefore, improving real-time temperature monitoring solutions will be crucial for more precise ATC implementation, particularly for WBG semiconductors, where faster switching speeds necessitate high-bandwidth temperature extraction methods.

Finally, overload management strategies utilizing ATC require further development to ensure reliable operation under transient conditions. Research should focus on developing adaptive ATC methods that apply light control actions during normal operation while reserving stronger interventions for extreme thermal stress scenarios.

VII. CONCLUSION

Despite significant advances in power electronics technologies, reliability remains a crucial concern for the successful deployment of these systems. The performance and longevity of power semiconductor devices, particularly Si-IGBTs and SiC-MOSFETs, are still affected by thermo-mechanical aging. Moreover, while IGBT devices benefit from established manufacturing processes and cost-effectiveness, SiC devices excel in high-temperature and high-voltage applications.

However, SiC devices are vulnerable to unique failure mechanisms, such as gate-oxide degradation and threshold voltage drift.

To address these limitations, this article has provided a comprehensive overview of reliability concerns across multiple levels of the power module design chain. First, at the reliability testing level, the main industrial standards (e.g., HTGB, HTRB, H3TRB, thermal cycling, power cycling) were reviewed. These tests were contextualized according to their objectives, typical conditions, and targeted failure mechanisms.

Moreover, this article explores the reliability of power semiconductor devices by integrating considerations at both the chip and package levels. At the chip level, the technological evolution of Si-IGBTs and SiC-MOSFETs was described in relation to how each generation addressed specific reliability challenges. In particular, the case of SiC-MOSFETs illustrates how early reliability barriers, such as poor oxide quality and high failure rates, led to the temporary adoption of alternative devices like SiC-JFETs or cascodes before trench-optimized MOSFETs became feasible. These improvements have allowed the development of increasingly robust devices capable of operating at higher temperatures and switching frequencies while reducing failure rates.

At the package level, key elements such as bond wires, die-attach materials, metallization, and substrates were analyzed. The discussion emphasized how each element contributes to thermal performance and lifetime. Aluminium wire bonding remains the most widely used method, yet its limited reliability under thermal stress has led to the development of copper wire, ribbon bonding, and wire-free alternatives. Similarly, the transition from solder-based die attach to silver sintering and advanced joining techniques has enabled higher operating temperatures and improved mechanical stability. These packaging innovations are crucial enablers for achieving longer module lifetimes under demanding conditions.

Regarding ATC strategies, this article has classified and assessed their role in extending the operational life of power semiconductors. These software- and hardware-based control methods were reviewed according to their implementation hierarchy, objectives, and effectiveness. Furthermore, ATC strategies were contextualized with respect to their compatibility with overload conditions, their validation through standard tests (e.g., SC or DPT), and their interaction with device limits such as SOA boundaries. From the user's perspective, ATC techniques such as adjusting gate voltage or switching frequency provide a practical and responsive means to mitigate temperature swings. Techniques like I_{load} limitation or V_{DC} regulation offer further protection mechanisms tailored to specific load and thermal conditions. While some strategies may affect signal quality or require hardware changes, the trade-off can be justified in critical applications.

As a result, the growing implementation of ATC strategies is expected to be particularly relevant for high-performance applications such as renewable energy systems, electric mobility, and high-power industrial drives, where reliability,

thermal stability, and extended lifetime are essential design objectives.

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ENEKO AGIRREZABALA received the B.Sc. degree in energy engineering, and the M.Sc. degree in energy and power electronics in 2019 and 2021, respectively from Mondragon Unibertsitatea (MU), Mondragón, Spain, where he has been working toward the Ph.D. degree, focusing on the implementation of active thermal control strategies to prolong lifetime and enhance overload capability in power semiconductor devices since 2023. Since 2021, he has been a Lecturer and a Researcher with the Electronics and Computing Department, Faculty of Engineering, MU. His research interests include power semiconductor reliability assessment, thermal modelling, and hardware design.



systems via power electronic converters. He has participated in various research projects in the fields of traction and stationary systems for railway applications, renewable energy applications and energy storage applications for traction, on-grid and off-grid systems.

IOSU AIZPURU received the B.Sc., M.Sc., and the Ph.D. in electrical engineering from the University of Mondragon, Mondragón, Spain, in 2006, 2009, and 2015, respectively. He is currently a Researcher and Lecturer with the Electronics and Computing Department, Faculty of Engineering, Mondragon Unibertsitatea, Mondragón. He is also the coordinator of Power Electronics Research Group, Valencia, Spain. His current research interests include power electronics modelling, energy storage modelling, and control of energy storage



converters, focusing on wide bandgap (WBG) devices and their application. He has participated in different industrial projects related to these fields.

DAVID GARRIDO received the degree in electronic technical engineering from Mondragon Unibertsitatea (MU), Mondragón, Spain, in 2004, the degree in electronic engineering from the Polytechnic University of Catalonia, Barcelona, Spain, in 2007, and the Ph.D. degree in electronic engineering from MU, in 2020. He joined Electronics and Computing Department, MU, in 2009, where he teaches subjects related to power electronics and electric vehicles. His main research interests include power semiconductor devices and power



the analysis of power semiconductors under various operating conditions, both experimentally and theoretically, with the aim of enhancing the reliability and efficiency of power converters through precise temperature management.

ANE PORTILLO received the B.Sc. degree in energy engineering from Mondragon Unibertsitatea (MU), Mondragón, Spain, in 2024, completing her final year with the Warsaw University of Technology, Warsaw, Poland. Her final degree project was focused on the optimization of the thermal modelling of power semiconductor modules. She is currently working toward the M.Sc. degree in smart energy systems with MU. She is currently a Research Assistant with the Electronics and Computing Department, MU. Her research focuses on